
Core Mathematics

A Practical Guide on Mathematics for Physics Learners

Sho Iwamoto

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Lecture notes for first-year university students in physics or physical science, covering mathematical tools you the students need most. Calculus, logic, complex numbers, and vectors, with introductions to vector calculus, differential equations, and statistical analysis. Exercises are central to these notes. Practice each topic repeatedly until you can work through problems quickly and accurately.

Preface

Physics is an activity to describe the world, but how? At present, mathematics (+ *broken* English) is the only language we can use. So, *unfortunately*, we need to learn mathematics! (and English... 🙄) But it means we are *fortunate*: we have a language to describe the world. In fact, physics have developed together with mathematics. We understand physics better if we study mathematics well!

When you do DIY, you need to use the tools properly, smoothly, and accurately. It is the same here; when we use math as a tool, it must be **logical**, **quick**, and **accurate**. To be logical, you must think each problem carefully; otherwise your discussion might be incomplete and you can't convince others. To be quick and accurate, you must "*drill*" repeatedly, as if top NBA players do shooting practice almost everyday. You should reach the point where basic calculations feel automatic.

This is not a math textbook. **This is rather** a math "drill" book. Rigorous proofs are mostly omitted, but drill problems are the core of this document, through which students are expected to achieve better conceptual understanding.

Target of this document

This document is primarily for first-year undergraduate students in their second semester, who

- completed basic calculus (differentiation and integration),
- are going to major in physics or physical science, and
- do not hesitate to do repetitive practice to achieve better conceptual understanding.

How to use this document

1. Buy a A4-sized paper notebook.

Digital notebooks on tablets are not suitable (less effective).

2. Read the text and solve **Quizzes**.

The texts are kept short, so please read all the text carefully. Quizzes should be solved while reading, but you don't have to write down the solution process for Quizzes.

3. Solve **Drill Problems**, until you feel "I can solve them *fluently and accurately*".

This is to improve your fluency ($\neq$ speed) and accuracy. Write down solution process clearly.

4. Solve other **Problems**.

This is to have better conceptual understanding, but also to develop your writing skills. Write down the solution process clearly *as an English text*, taking care of *logical completeness*.

Optionally, you are encouraged to prepare a reference book, so that you can consult it when you want to know more. Sho supposes [Boas] listed below is the best option.



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Visit <https://github.com/misho104/LecturePublic> for further information, updates, and to report issues.

Problems

-  **Drill problems**, marked with , are simple and easy. They help you practice the basic calculations until you can do them fluently and accurately. If they feel too easy for you, feel free to skip some of them.
- ★ These are **minimal basic** problems. To pass this course, you should be able to solve all of them. If you struggle with them, please come talk to Sho—don't wait until the exam!
 - *** These **basic** problems are the same difficulty as the ★-problems. Solve these for extra practice, to check that you really understand the ★-problems and were not just lucky.
 - ** These are **intermediate** problems. Solve these if you are aiming for an A+. Note also that, at this level, you are asked to write answers carefully with clear explanations.
 - * These are **advanced** problems, beyond the scope of the lecture, for motivated students who are interested in research activities, a top-level engineering career, or demanding analytical-thinking work.

Bibliography

This document is written under the influence of the following references.

- [The Math Book by Hal Tasaki](#) (数学：物理を学び楽しむために, in Japanese)
A free book with full description, but unfortunately in Japanese. Topic selections and the depth of descriptions are based on this reference.
- [First-Year Math Practice](#) (in Japanese)
This document is grounded in a “learn steadily” approach (“じっくり” in Japanese), which originates from Prof. Gocho and Prof. Kiyono, who were instructors of Sho in his freshman. Problems and rigorous math descriptions are taken from their works.
- **[Boas]** Mary L. Boas, *Mathematical Methods in the Physical Sciences*, 3rd ed., Wiley, 2006.
- **[AWH]** George B. Arfken, Hans J. Weber, and Frank E. Harris,
Mathematical Methods for Physicists, 7th ed., Academic Press, 2023.

These two books are widely used in physics departments around the world. [Boas] seems more friendly to beginners, while [AWH] seems more detailed and complete. Sho recommends you to study [Boas] *in parallel with* this document, using it as a reference, and to consult [AWH] when you want more details or more exercises.

Other references are:

- [1] Bureau International des Poids et Mesures, “The International System of Units (SI),” BIPM. [Online]. Available: <https://doi.org/10.59161/AUEZ1291>

Symbols

SI Units

Historically, different civilizations used different units. Today, science and engineering worldwide use the **SI units** (Système International d’Unités)—the single international standard. It is built from seven **SI base units**:

Table 1: SI base units and their symbols for dimension [1]. Recall that units are case-sensitive.

quantity	symbol and name	symbol for dimension
time	s (second)	T
length	m (meter)	L
mass	kg (kilogram)	M
electric current	A (ampere)	I
temperature	K (kelvin)	Θ
amount of substance	mol (mole)	N
luminous intensity	cd (candela)	J

Mathematical Notations

Chapter 1		Chapter 3	
$\mathbb{C}$	complex numbers	$\wedge$	logical “and”
$\mathbb{R}$	real numbers	$\vee$	logical “or”
$\mathbb{Q}$	rational numbers	$\neg$	logical “not”
$\mathbb{Z}$	integers	$\implies$	implication
$\mathbb{N}^0$	natural numbers (0, 1, 2, ...)	$\iff$	logical equivalence
$\mathbb{N}^+$	natural numbers (1, 2, ...)	$\forall$	for all
$x \in A$	a member of	$\exists$	there exists
$A \cup B$	union	$:=$	is defined as
$A \cap B$	intersection		
$A \setminus B$	set difference		
$f'(x)$	derivative $\left(\text{also } \frac{df(x)}{dx} \right)$		
$f^{(n)}(x)$	n -th derivative $\left(\text{also } \frac{d^n f(x)}{dx^n} \right)$		

Greek symbols

A	α	alpha	H	η	eta	N	ν	nu	T	τ	tau
B	β	beta	Θ	θ	theta	Ξ	ξ	xi	Y	υ	upsilon
Γ	γ	gamma	I	ι	iota	O	$\omicron$	omicron	Φ	$\varphi(\phi)$	phi
Δ	δ	delta	K	κ	kappa	Π	π	pi	X	χ	chi
E	ε	epsilon	Λ	λ	lambda	P	ρ	rho	Ψ	ψ	psi
Z	ζ	zeta	M	μ	mu	Σ	σ	sigma	Ω	ω	omega

Gray-outed ones are almost never used, but others are almost always used. You need to write them so that **others can distinguish each from others**, but Sho thinks Japanese- or Chinese-speakers are OK with it, as we everyday read and write 日 vs 日 vs 白 or 土 vs 土 vs 工 properly!

Quiz

Q1. Read out the following words.

- | | | |
|---|----------------------------------|---|
| (1) α -particle | (2) β -decay | (3) Γ -function |
| (4) ε - δ definition | (5) angle θ and φ | (6) μ - and τ -leptons |
| (7) $\psi(x)$ and $\varphi(x)$ | (8) ζ and ξ | (9) metric $\eta_{\mu\nu}$ |
| (10) X-ray and γ -ray | (11) Ω -baryon | (12) $\varepsilon_{\mu\nu\rho\sigma}$ -tensor |

Problems

 0.1 Write the following letters so that people can distinguish each from others.

- | | | | |
|--------------------|------------------------------------|---------------------|-------------------------|
| (1) a, α | (2) b, β | (3) c, C | (4) e, E, ε |
| (5) ζ, ξ | (6) $\Theta, \theta, Q, \sigma, 6$ | (7) $g, q, 9$ | (8) $i, l, 1, I$ |
| (9) k, K, κ | (10) m, μ | (11) p, ρ | (12) s, S |
| (13) t, τ, T | (14) u, v, ν | (15) w, W, ω | (16) x, X, χ |

Lecture Plan

The content is designed for 150-minute $\times$ 14-week lectures, as it is originally for a lecture *Mathematics for Fundamental Physics* (基礎物理數學) in National Sun Yat-sen University (國立中山大學).

1. Derivative
2. Units. Significant Digits.
3. Logic. Theorems. a^x and $\log_a x$.
4. Logic. Theorems. a^x and $\log_a x$.
5. Vectors are arrows.
6. $a \cdot b$ and $a \times b$
7. Matrices (rotation, reflection, scaling)
8. Complex numbers (polar form, Euler's formula, multi-valuedness)
9. Complex vectors, Hermitian inner product, bra-ket notation
10. Vector calculus (grad, div, rot; example of point charge)
11. ODE basics
12. ODE basics
13. Probability and error analysis
14. Probability and error analysis

♣TODO♣ a bit on (abstract) vector space to prep for QM?

Contents

Preface	2
Bibliography	3
Symbols	4
Lecture Plan	6
Index	10
1 Derivative (Review)	11
1.1 The first step	11
1.2 Typical Pitfalls	12
1.3 Taylor expansion	14
1.4 Mathematical Notation	16
2 Units and Significant figures	18
2.1 Units	19
2.2 Significant figures	21
2.3 Calculation with Significant figures	22
3 Logic	25
3.1 Basic Logics	25
3.2 Implication $\Rightarrow$	26
3.3 Equivalent Statements—What does “solve” mean?	27
3.4 A few more notes about logic and proof	31
3.5 Facts, Definitions, Assumptions, and Conclusions	32
4 Power, Exponential, and Logarithm	35
4.1 Powers	35
4.2 Exponential functions and Logarithmic functions	38
4.3 Napier’s number and Natural logarithm	39
4.4 Exponential Growth and Decay	41
5 Vectors as arrows	43
5.1 What is a Vector?	43
5.2 Addition and Scalar multiplication	44
5.3 Inner Product	46
5.4 Cross Product (only for 3d-vectors)	48
5.5 Position Vectors	49
5.6 Intermission: Vector or Scalar or Not . .	50
5.7 Axes and Components	51
6 Complex Numbers	55
6.1 Complex Numbers	55
6.2 Polar Form	60
6.3 Euler’s formula	61
7 Matrices	65

List of Theorems and Definitions

Example 1.1	 14	Definition 6.8	... 61
Statement 1.2	 15	Theorem 6.9	 61
Statement 2.1	 19	Theorem 6.10	... 62
Example 2.2	 23	Example 6.11	 62
Example 2.3	 23	Example 6.12	 63
Example 3.1	 26	Example 6.13	 63
Definition 3.2	... 27		
Theorem 3.3	 31		
Example 3.4	 31		
Example 3.5	 32		
Definition 4.1	... 36		
Example 4.2	 36		
Example 4.3	 36		
Example 4.4	 36		
Example 4.5	 37		
Definition 4.6	... 37		
Definition 4.7	... 37		
Theorem 4.8	 38		
Definition 4.9	... 38		
Theorem 4.10	... 39		
Theorem 4.11	... 39		
Definition 5.1	... 43		
Notation 5.2	 44		
Definition 5.3	... 44		
Definition 5.4	... 45		
Definition 5.5	... 45		
Example 5.6	 45		
Theorem 5.7	 46		
Definition 5.8	... 47		
Theorem 5.9	 47		
Definition 5.10	.. 48		
Theorem 5.11	... 49		
Definition 5.12	.. 49		
Definition 5.13	.. 51		
Example 5.14	 51		
Theorem 5.15	... 52		
Definition 5.16	.. 52		
Example 5.17	 53		
Definition 6.1	... 55		
Definition 6.2	... 55		
Definition 6.3	... 56		
Theorem 6.4	 58		
Example 6.5	 58		
Theorem 6.6	 60		
Definition 6.7	... 60		

List of tables

Table 1 SI base units and their symbols for dimension [1]. Recall that units are case-sensitive. . . 4

List of figures

Index

n -dimensional space	53	half-life	41	rational number	16
n -dimensional vector	53	hyperbolic function	41	real part	55
absolute uncertainty	18	iff	28	relative uncertainty	18
absolute value	44 , 56	imaginary part	55	right-hand rule	48 , 52
addition	44	imaginary unit	55	right-handed system	49 , 52
and operator	26	implication	27	roots of unity	64
anti-parallel	45	infinite descent	32	rounding	22
argument	60	inner product	47	scalar	44
axis	52	integer	16	scalar multiplication	44
base	35	intersection	17	scientific notation	21
Cartesian coordinate system		inverse	31	set	17
	49 , 52	logarithm	38	set difference	17
central value	18	magnitude	43	SI base unit	4
complex conjugate	56	mathematical induction	32	SI unit	4
complex number	55	multivalued	60	significant figure	18 , 21
complex plane	55	Napier's number	39	sufficient condition	28
conjunction	→ see <i>and</i>	natural logarithm	39	syllogism	34
constant	12	necessary condition	28	symbols for dimension	20
contrapositive	31	negation	→ see <i>not</i>	tangent line	14
converse	31	normal	47	component	52
coordinate system	52	not operator	26	transfinite induction	32
counterexample	26	or operator	26	triangle inequality	46
cross product	48	order	13	trigonometric function	14
de Moivre's theorem	62	orthonormal basis	51	trigonometric function	60
de Morgan's theorem	30 , 31	parallel	45	true	25
definition	14	perpendicular	47	truth table	26
derivative	11	physical quantity	18	uncertainty	18
dimension	52	polar form	60	absolute	18
dimensionless	19	position vector	49	relative	18
direction	43	power	35	union	17
disjunction	→ see <i>or</i>	power serie	60	unit	18
e	→ see <i>Napier's number</i>	principle value	60	unit vector	46
empty set	17	proof	25	variable	12
equivalent	27	proof by contradiction	32	vector	43
error	→ see <i>uncertainty</i>	proof by contrapositive	31	zero vector	44
Euler's formula	61	proposition	34		
exponent	35	pure-imaginary	55		
exponential decay	41	quantifier	30		
exponential function	38	radian	13		
exponential growth	41	rate of change	14		
false	25				

Chapter 1

Derivative (Review)

1.1 The first step

Birds sing, fish swim, flowers bloom, stars twinkle, and university students calculate derivatives. Let us begin with basic **derivatives**. Try the next quiz—and note how long it takes.

Quiz

Q1. Calculate the first derivatives of the following functions, **measuring how many minutes it takes**.

(1) $(x + 1)^3$

(2) $\tan x$

(3) $2 \cos^2 x$

(4) $2 \cos x^2$

(5) $x^3 \cos x$

(6) $\frac{3 \sin x}{x}$

(7) $x^{3/2}$

(8) $\sqrt{\sin x}$

- Less than two minutes? Amazing! It is as fast as Sho!
- Less than four minutes? Great, it is exactly as Sho anticipates. Please continue your effort!
- Even if you took more than four minutes, do not worry. At least you don't make any mistakes. A little more practice will help you. Try more problems in [the Derivative Boot Camp](#).^{#1}
- If you make any mistakes in these eight calculations, then it is a serious issue—just as serious as forgetting how to do the calculations at all. In university, you will perform similar calculations more than 100 times. You need to be both fast and accurate.



At university, students often underestimate the importance of basic calculations. In physics, simple calculations appear constantly, so your speed and accuracy directly affect how well you follow lectures, how efficiently you study, and ultimately your grade. Both can be improved with practice.

This course will help you strengthen these basic calculations—some of which you already know from high school—while also introducing new topics that are essential for physics. You will solve many problems and drills, just as an athlete repeats the same move hundreds of times to master it.

This document (this lecture course) is designed for first-year students in their second semester, **who have already studied derivatives well** in their first-semester calculus. As we do not cover the details of derivatives here, if you are not confident with derivatives, please first study [the Derivative Boot Camp](#).

^{#1}Visit <https://misho104.github.io/LecturePublic> and find gp1_boot1_deriv_true.pdf.

1.2 Typical Pitfalls

1.2.1 Notations

Consider a function $f(x)$. As we vary the value of x , x is called a **variable**. For a function $f(x)$, there are several equivalent ways to write its derivative:

$$f'(x) = \frac{df}{dx}(x) = \frac{df(x)}{dx} = \frac{d}{dx}f(x) \quad [\text{All means the same thing}]. \quad (1.1)$$

Similarly, the value of $f'(x)$ at a specific point $x = 3$ can be written as

$$f'(3) = f'(x)\Big|_{x=3} = \frac{df}{dx}\Big|_{x=3} = \frac{df}{dx}(3) = \frac{d}{dx}f(3), \quad (1.2)$$

and they are all equivalent. For example, if $f(x) = 3x^2 + 6$, then $f'(x) = 6x$ and $f'(3) = 18$.

Most of students get confused when a **constant** a appears. If a is *declared as a constant*, we can define a function such as $g(x) = ax^2 + 2a$, for which

$$g'(x) = 2ax. \quad (1.3)$$

We can evaluate $g'(x)$ at $x = a$. The result is $g'(a) = 2a^2$, written as

$$g'(a) = g'(x)\Big|_{x=a} = \frac{dg}{dx}(a) = \frac{d}{dx}g(a) = 2a^2. \quad (1.4)$$

Be careful: The notation $\frac{d}{dx}g(a)$ does **not** mean $\frac{d}{dx}[g(a)] = \frac{d}{dx}(a^3 + 2a) = 0$.

Fail safe note: If you can't see Eq. (1.3), try setting $a = 3$. Then, you notice $g(x)$ is the same function as $f(x)$, and $g'(x)$ should equal $f'(x)$.

Quiz

Q2. Let a be a constant and $g(x) = ax^2 + 2a$. Calculate

- | | | | |
|-------------|-------------|--------------|------------------------|
| (1) $g(1)$ | (2) $g(a)$ | (3) $g(0)$ | (4) $g(7)$ |
| (5) $g'(1)$ | (6) $g'(a)$ | (7) $g''(2)$ | (8) $\frac{d}{dx}g(7)$ |

One more bad news. *Physicists are often lazy* and write $f(x)$ as f . If $f(x) = 2x^2 + 1$, we may write $f' = 4x$ and $f'' = 4$. For $g(t) = 2t^2 - 1$, you may write $g' = 4t$ and $g'' = 4$. Therefore, when you see a function f , you must **identify its variable from the context**.

In physics, a variable can be a function of another variable. The kinetic energy $K(v) = mv^2/2$ is a good example. Its variable v is a function of time t —we write this fact by $v = v(t)$ —and thus

$$K(v) = \frac{1}{2}mv^2 \quad \text{but also} \quad K(t) = K(v(t)) = \frac{1}{2}mv(t)^2. \quad (1.5)$$

Now, what does K' mean? —It is ambiguous and we must avoid such notation. Even though, if you see it in a textbook, you need to guess the author's intention from the context.

Advanced note: This rewrite, $K(t)$ and $K(v)$, is common in physics. However, mathematicians do not like this practice because they consider K a unique object: a function should always mean the same rule applied to its input. If they see $K(v) = mv^2/2$, they would say [K is an object that converts its input into $m/2 \times (\text{input})^2$, and thus if you feed t into K , you must get $K(t) = mt^2/2$]. This is of course not what we mean. For example, imagine $v(t) = \alpha t$ with α a constant. Then

$$\text{physicists: } K(v) = \frac{1}{2}mv^2, \quad K(t) = \frac{1}{2}m\alpha^2 t^2; \quad (1.6)$$

but mathematicians would say, [since it returns $m\alpha^2/2 \times (\text{input})^2$, it is a different object] and hence use a different name for it, such as $\tilde{K}$:

$$\text{mathematicians: } K(v(t)) = \frac{1}{2}m \times v(t)^2 = \tilde{K}(t) = \frac{1}{2}m\alpha^2 t^2. \quad (1.7)$$

Both conventions are reasonable, and eventually you will get used to both.

Higher-order derivatives is written as

$$\frac{d}{dx} \left(\frac{df}{dx} \right) = \frac{d^2 f}{dx^2} = f''(x) = f^{(2)}(x), \quad \frac{d}{dx} \left(\frac{d}{dx} \left(\frac{df}{dx} \right) \right) = \frac{d^3 f}{dx^3} = f'''(x) = f^{(3)}(x), \quad (1.8)$$

etc. Please be careful on the position of “2” and “3”.

1.2.2 Radian and trigonometric functions

At university, angles are almost always measured in **radians**:

$$360 \text{ degree (or: } 360^\circ) = 2\pi \text{ radian (or: } 2\pi \text{ rad)} \quad (1.9)$$

Furthermore, we usually omit “radian” (because we are lazy!). So,

$$\text{a right angle is } \pi/2. \quad \text{The sum of the interior angles of a triangle is } \pi. \quad (1.10)$$

Quiz

Q3. Express the following angles in radians, and radians in angles.

- | | | | | |
|-----------------|----------------|----------------|---------------|-----------------|
| (1) 180° | (2) 45° | (3) 60° | (4) 90 | (5) -30° |
| (6) 1° | (7) 1 | (8) -0.3 | (9) x° | (10) x |

Why do insist on radians? The answer comes from the derivative formula

$$\frac{d}{dx} \sin x = \cos x. \quad (1.11)$$

Because we have chosen radians as the standard, $(\sin x)'$ becomes this simple.

Advanced note: The simpleness of Eq. (1.11) originates in the fact $\sin(0.01 \text{ rad}) \approx 0.01$. If we used degrees, we would have $\sin(0.01^\circ) \approx 0.01 \times 0.017453$ and everything is messed up with this number 0.017453.

There are a few remarks in the notation of **trigonometric functions**:

$$\begin{aligned} \sin^2 x &\neq \sin x^2. & \text{Namely, } (\sin x)^2 = \sin^2 x &\neq \sin x^2 = \sin(x^2). \\ \tan^{-1} x &\neq \frac{1}{\tan x}. & \text{Namely, } \tan^{-1} x = \arctan x &\neq (\tan x)^{-1} = \frac{1}{\tan x} = \cot x. \end{aligned} \quad (1.12)$$

The following expressions are not incorrect but confusing;

$$\sin^{-2} x, \quad \sin^{1/2} x, \quad \sin(x)^2, \quad \sin(x+1)^2, \quad \dots$$

We should avoid ambiguity, so please *never* use confusing these notations. Sho thinks we should use $\sin^k x$ only for $k = 2, 3, 4, \dots$, and use $\arcsin x$ instead of $\sin^{-1} x$.

1.2.3 Several interpretations of derivatives

When you, physics learners, discuss $f'(t)$, you should have the following three interpretations:

- Regarding t as the time, $f'(t)$ is the **rate of change** of $f(t)$ per unit time. If $f'(t) > 0$, the function f is increasing at the time t , while $f'(t) < 0$ means it is decreasing.
- Consider a graph of $f(t)$. Then, $f'(t)$ is the slope of the **tangent line** to the graph at the point t .
- Mathematically, $f'(t)$ is defined by (Check that they are equivalent.)

$$f'(t) := \lim_{\Delta t \rightarrow 0} \frac{f(t + \Delta t) - f(t)}{\Delta t} = \lim_{h \rightarrow 0} \frac{f(t + h) - f(t - h)}{2h} = \lim_{s \rightarrow t} \frac{f(s) - f(t)}{s - t}. \quad (1.13)$$

If you are unsure of them, please consult your first-year Calculus textbooks for further information.

Quiz

Q4. What is the definition of $f'(x)$? Explain.

In physics, it is important to memorize and understand the **definition** of each concept.

1.3 Taylor expansion

Reviewing the definition of $f'(x)$, we have

$$f'(x) := \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}, \quad f'(a) := \lim_{\delta \rightarrow 0} \frac{f(a + \delta) - f(a)}{\delta}. \quad (1.14)$$

The second equation is interpreted as follows:

$$\text{If } \delta \approx 0, \quad f'(a) \approx \frac{f(a + \delta) - f(a)}{\delta}, \quad \text{i.e., } f(a + \delta) \approx f(a) + \delta f'(a) \quad (1.15)$$

and this interpretation is useful in the following example:

Example 1.1

Find the approximate value of the following expressions without using calculators. Then, check your answer with a calculator.

1. $\sqrt{1.002}$

2. $(1.002)^{10}$

3. $\sqrt{4.004}$

Solution

(1) Apply Eq. (1.15) for $f(x) = \sqrt{x}$, $a = 1$, and $\delta = 0.002$. Then,

$$f'(x) = \frac{1}{2\sqrt{x}}, \quad f'(a) = f'(1) = \frac{1}{2}, \quad f(a + \delta) \approx f(a) + \delta f'(a) = 1 + \frac{\delta}{2} = \underline{1.001}.$$

(2) Doing the same thing for $g(x) = x^{10}$ with $a = 1$ and $\delta = 0.002$,

$$g'(x) = 10x^9, \quad g'(a) = g'(1) = 10, \quad g(a + \delta) \approx g(a) + \delta g'(a) = 1^{10} + 10\delta = \underline{1.02}.$$

(3) Doing the same thing for $h(x) = \sqrt{x}$, $a = 4$, and $\delta = 0.004$, we have

$$h'(4) = \frac{1}{4}, \quad h'(4.004) \approx h(4) + 0.004 \times \frac{1}{4} = \underline{2.001}.$$

Advanced note: “Solutions” contain not only the answer but also how you reached the answer. You are, of course, asked to write such explanations when you solve problems.

Namely, with this technique, you can find **the value of $f(x)$ around a point $x = a$** . This technique, Eq. (1.15), is a special case of Taylor’s theorem, which is discussed in **♣TODO♣???**.

Quiz

Q5. Find the approximate value of the following expressions without using calculators. Then, check your answer with a calculator.

(1) $\sqrt{1.001}$

(2) $(1.0001)^{30}$

(3) $\frac{1}{1.001}$

To summarize, we have the following statement:

Statement 1.2 (Taylor expansion: basic)

For a physicists-friendly function $f(x)$,

$$f(a + \delta) \approx f(a) + \delta f'(a), \tag{1.16}$$

or as an equivalent expression,

$$f(x) \approx f(x_0) + (x - x_0)f'(x_0). \tag{1.17}$$

Quiz

Q6. Check these two equations are equivalent. You will be ready to use either of them in the future.

Q7. Review how to derive Eq. (1.16) and summarize it in your own words.

Problems

 **1.1** Calculate the following expression.

- (1) $\sin 30^\circ$ (2) $\cos 45^\circ$ (3) $\cos 120^\circ$ (4) $\sin 150^\circ$ (5) $\tan 180^\circ$
 (6) $\cos 0$ (7) $\sin 2\pi$ (8) $\cos \pi$ (9) $\sin(\pi/3)$ (10) $\cos(-\pi/4)$
 (11) $\tan(2\pi/3)$ (12) $\sin(\pi/6)$ (13) $\tan(5\pi/6)$ (14) $\cos(-\pi/6)$ (15) $\tan(-\pi/2)$

 **1.2** For the following $f(x)$, calculate $f'(x)$, $f''(x)$, $f^{(3)}(x)$, $f^{(4)}(x)$, and $f^{(5)}(x)$.

- (1) x^3 (2) $\sin 2x$ (3) e^{2x} (4) $\ln 2x$

 **1.3** Find the approximate value for the following expressions without calculators. Then, check your answer with a calculator.

- (1) $(1.002)^7$ (2) $(2.002)^7$ (3) $(0.998)^4$ (4) $(10.01)^3$ (5) $1 \div 0.998$
 (6) $\sqrt{0.998}$ (7) $\sqrt{4.001}$ (8) $(0.999)^{1/2}$ (9) $(0.999)^{-1/2}$ (10) $(0.999)^{-3/2}$
 (11) $\frac{1}{(2.02)^2}$ (12) $\tan 0.001$ (13) $\cos 0.002$ (14) $\sin 0.003$ (15) $\ln 1.001$

****1.4** Find the approximate value for the following expressions without calculators. Then, check your answer with a calculator.

- (1) $\sqrt[3]{1.001}$ (2) $\sqrt[3]{8.012}$ (3) $1.001^{0.5}$ (4) $1.001^{-0.9}$ (5) $\sin(3.14)$
 (6) $\cos(1.57)$ (7) $\sin 1^\circ$ (8) $\cos 61^\circ$ (9) $\ln 0.999$ (10) $\log_{10}(10.01)$
 (11) $\sqrt{1.001} \sin 0.001$

Fail safe note: If you are not sure about $\sqrt[3]{1.001}$, $(1.001)^{-0.9}$, etc., study [Chapter 4](#) first.

****1.5** For the following $f(x)$, calculate $f^{(n)}(x)$ for all positive integers $n = 1, 2, 3, \dots$

- (1) x^{10} (2) $\sin 2x$ (3) e^{2x} (4) $\ln x^2$ (5) $\sqrt{x}$

***1.6** Write down the definition of the second derivative $f''(x)$. Repeat the discussion of Eq. (1.15) to find the expansion $f'(a + \varepsilon) \approx f'(a) + \varepsilon f''(a) + (\varepsilon^2/2)f''(a)$.

Advanced note: The two equations in Eq. (1.14) have similar but different meanings: the first one defines a new function $f'(x)$, while the second one defines a number that is eventually equal to $f'(x)|_{x=a}$. Anyway, we don't care the difference.

1.4 Mathematical Notation

As *physicists are lazy*, we usually use the following notation:

$x \in \mathbb{C}$	means “ x is a complex number”.	$x \in \mathbb{Z}$	means “ x is an integer ”. ^{#2}
$x \in \mathbb{R}$	means “ x is a real number”.	$x \in \mathbb{N}^+$	means “ x is a positive integer”.
$x \in \mathbb{Q}$	means “ x is a rational number ”.	$x \in \mathbb{N}^0$	means “ x is a non-negative integer”.

^{#2} $\mathbb{Z}$ comes from “Zahlen” (Zahl).

The symbol “ $\in$ ” means “*is a member of*”, and $\mathbb{N}^+$ means a **set** of positive integers, $\{1, 2, 3, 4, \dots\}$. So,

$x \in \mathbb{N}^+$ and $x \in \{1, 2, 3, 4, 5, \dots\}$ means the same thing: “ x is a positive integer”.

Remark: In mathematics, “natural numbers” usually mean $\mathbb{N}^0$. Meanwhile, physicists and other people tend to think natural numbers mean $\mathbb{N}^+$.

Quiz

Q8. What is the difference between $\mathbb{N}^+$ and $\mathbb{N}^0$?

Q9. What is the difference between $\mathbb{N}^0$ and $\mathbb{Z}$?

Q10. What is “rational numbers”? What is the difference between $\mathbb{Z}$ and $\mathbb{Q}$?

We write “ x is a positive number” by $x \in \mathbb{R}, x > 0$ but also by $x > 0$. Similarly,

$x \in \mathbb{R}, x \leq 0$ or $x \leq 0$ means “ x is a non-positive number”.

Notice that, when we say “positive” or “negative”, we implicitly assume the number is real.

Advanced note: This is because we cannot compare complex numbers with 0. The operator $>$ is defined only for real numbers. When we write $a > b$, we implicitly consider that a and b are real numbers.

Problems

****1.7** The **intersection** of two sets A and B is a set made by elements both in A and B ; we write it by $A \cap B$. The **union** of A and B is a set made by elements in A , B , or both; we write it by $A \cup B$. The **set difference**, $A \setminus B$, is a set made by elements in A but not in B .

For example, let $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, and $C = \{2, 4\}$. Then,

$$A \cap B = \{2, 3\}, \quad A \cup B = \{1, 2, 3, 4\}, \quad A \setminus C = \{1, 3\}, \quad B \setminus C = \{3\}, \quad C \setminus B = \{\},$$

where $\{\}$ (or $\emptyset$) means the **empty set**, the set without any elements.

(1) Write the following sets.

- (a) $A \cap C$ (b) $B \cap C$ (c) $A \cup C$ (d) $B \cup C$ (e) $A \cup B \cup C$
 (f) $B \setminus A$ (g) $C \setminus A$ (h) $C \setminus B$ (i) $(B \setminus A) \cup A$ (j) $\mathbb{N}^0 \setminus \mathbb{N}^+$

(2) Express the following statements only with $\mathbb{R}$, $\mathbb{Q}$, $\mathbb{Z}$, $\mathbb{N}^+$, $\mathbb{N}^0$, $\cap$, $\cup$, $\setminus$, and “ $x \in$ ”.

- (a) x is a rational number. (b) x is an irrational number.
 (c) x is a rational number but not an integer. (d) x is a non-positive integer.
 (e) x is a negative integer. (f) x is a non-zero integer.

Chapter 2

Units and Significant figures

A **physical quantity** is usually not just a number, but also has a **unit** and an **uncertainty** (error). For example, “ $(72 \pm 2) \text{ kg}$ ” has a number “72” and a unit “kg”, which compose the **central value** “72 kg”. It also has uncertainty because every measurement has an uncertainty. In experimental physics, uncertainty is usually more important than the central value, and thus uncertainty handling is a fundamental skill of physicists.

In this chapter, we first review physical quantities and how to handle units. Then, we discuss **significant figures**, a simple and convenient method to express the uncertainty. Further discussions on uncertainty analysis are given in Chapter ♣TODO♣chapter.

=s Physical quantity

The expression “ $(72 \pm 2) \text{ kg}$ ” means the central value is 72 kg and the **absolute uncertainty** is 2 kg. Accordingly, its **relative uncertainty** is given by $(2 \text{ kg})/(72 \text{ kg}) = 0.028$. Let’s see other examples.

	unit	central value	absolute uncertainty	relative uncertainty
$10 \text{ m} \pm 1 \text{ cm}$	m (meter)	10 m	1 cm = 0.01 m	0.001 (or 0.1%)
$1.6 \text{ A} \pm 0.04 \text{ A}$	A (ampere)	1.6 A	0.04 A	0.025 (or 2.5%)
$(5 \pm 1) \times 10^{-3} \text{ C}$	C (coulomb)	0.005 C	0.001 C	0.2 (or 20%)
$(50 \pm 1) \text{ m/s}$	m/s	50 m/s	1 m/s	0.02 (or 2%)

Remark: Upper- and lowercase letters are distinguished. “M” **does not** mean “meter”. The unit “A” is “ampere”, not “Ampere”. The unit “coulomb” is “C”, not “c”.

Quiz

Q1. For each of the following values, find its unit, central value, absolute uncertainty, and relative uncertainty.

(1) $50 \text{ kg} \pm 500 \text{ g}$

(2) $(0.05 \pm 0.001) \text{ A}$

(3) $(1.6 \pm 0.001) \times 10^{-19} \text{ C}$

(4) $72 \text{ km/h} \pm 1 \text{ m/s}$

Q2. The next passage has six (6) errors in the use of uppercase and lowercase letters. Find them out.

In the SI system, temperature is expressed in k (kelvin), a unit named after Lord Kelvin. The unit of force is N (newton), named after the british scientist Isaac newton. The units A (Ampere) and c (coulomb) are named after French scientists. In contrast, Kg (kilogram) is not named after a person. , but comes from Greek.

As an undergraduate student, you need to follow the following rules:

Statement 2.1 (Rules for Physical Quantities: Basic)

1. Numbers are always with units, even in calculations.
2. Include units in a symbol.
3. For physical quantities, use decimals (e.g., 1.6 A). Do not use fractions like (8/5) A.
4. Use significant figures to express the measurement precision.

Be careful: We use *decimals for physical quantities* and *fractions for mathematical concepts*. Since this lecture is mainly on math and math concepts, you will use decimals **♣TODO♣only in this section.**

2.1 Units

When we educate kids, we write “my height is $\square$ cm”, or “my height is h cm”, where $h = 172$ is just a number. But this is not nice! We want to convert the units freely and write equations such as $1.72 \text{ m} = 172 \text{ cm} = 0.00172 \text{ km}$. So, **we always include units in symbols** $h = 172 \text{ cm}$. Then

we can write: $h = 172 \text{ cm} = 1.72 \text{ m} = 0.00172 \text{ km} = 1.82 \times 10^{-16} \text{ light-year}$.

Similarly, if $m = 110 \text{ g}$ and $g = 9.8 \text{ m/s}^2$,

we may write: $w = mg = 110 \text{ g} \times 9.8 \text{ m/s}^2 = 1.1 \text{ kg} \cdot \text{m/s}^2$
 but not: $w = mg = 0.11 \times 9.8 = 1.1 \text{ kg} \cdot \text{m/s}^2$

This second equation is incorrect because m is not equal to 0.11; m is equal to 0.11 kg or 110 g.

Remark: Usually, physicists use upright fonts for units and *italic fonts* for quantities. For example, m , T , and C are quantities, which can be mass, temperature, capacitance, etc. Meanwhile, m, T, and C are units: meter, tesla, and coulomb, respectively.

Quiz

- Q3.** (1) If $m r \omega^2 = 10.0 \text{ kg} \cdot \text{m/s}^2$, $m = 5.0 \text{ kg}$, and $r = 1.0 \text{ m}$, what is ω ?
- (2) If $v_0 t + \frac{1}{2} a t^2 = 5.0 \text{ m}$, where $a = -4.0 \text{ m/s}^2$ and $v_0 = 7.0 \text{ m/s}$, what is t ?
- (3) At time $t = 0$, a particle is at $(x, y) = (2.0 \text{ m}, 0)$. It moves with a constant velocity $(v_x, v_y) = (3.0, 4.0) \text{ m/s}$. What is its position at $t = 1.0 \text{ s}$?

Every physical concept has its own unit. For example, speed has m/s, acceleration has m/s^2 , and energy has $\text{kg} \cdot \text{m/s}^2 = \text{N} \cdot \text{m} = \text{J}$. The table below lists the quantities you have learned. Notice that angle (rad) has no dimension. It is called a **dimensionless** quantity.

concept	dimension	typical unit
speed	L T^{-1}	m/s
acceleration	L T^{-2}	m/s^2

concept	dimension	typical unit
linear momentum	$M L T^{-1}$	$kg \cdot m/s$
force	$M L T^{-2}$	$kg \cdot m/s^2$ (= N)
energy / work	$M L^2 T^{-2}$	$kg \cdot m^2/s^2$ (= J)
frequency	T^{-1}	$1/s$ (= Hz)
angle	1 (dimensionless)	rad

Advanced note: To understand why angle has no dimension, you may consider the definition of the radian: it is defined as the ratio of the arc length to the radius, so the units cancel (meter / meter = 1).

Here, please do not write: The force is $kg \cdot m^2/s^2$. This is also called newton (N)., because we cannot mix units and concepts. A correct (and easy) way is to use English words:

the unit of the force is $kg \cdot m/s^2$ or the force has the unit $kg \cdot m/s^2$.

Similarly, the following statement is incorrect because it mixes concepts and units:

Since $m = kg$ and $g = m/s^2$, $mg = kg \cdot m/s^2 = N$.

Instead, you need to write

Since m has a unit of kg and g has a unit of m/s^2 , mg has a unit of $kg \cdot m/s^2 = N$.

Notice the last equation: units can be equal to other units, so we may write $N = J/m = kg \cdot m/s^2$
 $m = J/N$ $s = \sqrt{kg \cdot m/N}$. These are scientifically correct equations.

Remark: We sometimes use informal notations, such as $[F] = N$ or $F \sim N$ to express “the unit of F is N (newton)”. Still, using an equal symbol (=) is **not allowed**. Namely, $F = N$ is always incorrect.

Quiz

Q4. What are the units of the following concepts? Write English sentences to explain it.

(1) speed (2) velocity (3) force (4) energy (5) power

Advanced note: In very formal situations, we use **symbols for dimensions** (see Table 1 on page 4):

Since $\dim(m) = M$ and $\dim(g) = L T^{-2}$, $\dim(mg) = M L T^{-2}$, with the operator “dim” [1]. However, it seems too complicated for most situations.

Problems

★2.1 Express the following units only with SI base units, e.g., $N = kg \cdot m/s^2$.

- (1) W (watt) (2) Pa (pascal) (3) J (joule) (4) C (coulomb)

***2.2 Find a few more examples of dimensionless quantities in physics.

**2.3 Use “dim” notation to express the dimensions of the following quantities. For example, if v is speed, then $\dim(v) = L T^{-1}$.

- | | |
|--------------------------|--------------------------------------|
| (1) speed v | (2) force F |
| (3) angular momentum L | (4) electric charge Q |
| (5) resistance R | (6) specific heat capacity c |
| (7) Planck constant h | (8) fine-structure constant α |

2.2 Significant figures

In researches, we need to treat uncertainties in the method given in [♣TODO♣chap](#). However, most of lectures use **significant figures** to express the accuracy of a value, so that students becomes familiar with the concept of uncertainties. For example,

3.14	means the last digit “4” is uncertain. It can be 3.16, 3.15, 3.11, 3.10,
3.141	means “3.14” is for sure but it can be 3.143, 3.142, 3.140, 3.139,
3.1415	might be 3.1416, 3.14153, etc., but the writer is sure <i>it is not</i> 3.145 or 3.135.

We write the “uncertain digit” in [a different style](#) for clarity. You should notice the difference between:

1.0000	meaning the last digit “0” is uncertain so it may be 1.0002, 1.0001, 0.9998,
1.00	meaning it can be 1.03 or 0.99, but the writer is sure <i>it is not</i> 1.2 or 0.8.

It is usually convenient to use **scientific notation**:

0.000123	is OK but we prefer	1.23×10^{-4} .	Both mean the value can be 1.22×10^{-4} etc.
0.00100	is OK but we prefer	1.00×10^{-3} .	
29979	is OK but we prefer	2.9979×10^4 .	

Fail safe note: $10^3 = 1000$ and $10^{-3} = 1/10^3 = 1 \div 1000$. Go to [♣TODO♣chap](#) for details!

We need to *avoid ambiguous notations*. Namely,

2040	is ambiguous and not nice. We are not sure the author means 2.04×10^3 or 2.040×10^3 .
120	is not nice. We are not sure the author means It may mean 1.20×10^2 or 1.2×10^2 .
42000	is not nice because it has four ways to interpret the author’s intention.

Quiz

Q5. What are the four interpretations of 42000? Write them in scientific notation.

We use **rounding-to-the-nearest** (四捨五入) when necessary. For example, if you need to convert to three significant figures, it will be

$1.234 \rightarrow 1.23$, $1.235 \rightarrow 1.24$, $31.98 \rightarrow 32.0$, $1234 \rightarrow 1.23 \times 10^3$, and so on.

Quiz

Q6. Round the following numbers to three significant figures.

- | | | | |
|------------|---------------|------------|-------------------------|
| (1) 41.11 | (2) 98.76 | (3) 100.12 | (4) 15.449 |
| (5) 2.2360 | (6) 0.0123456 | (7) 9876 | (8) 9.999×10^4 |

Remark: Programmers use 2.99e8 (or 2.99E8) to mean 2.99×10^8 , or $1.6\text{e-}12$ (or $1.6\text{E-}12$) to mean 1.6×10^{-12} . However, we **should not** use them in handwriting.

Problems

 2.4 Write the following numbers in scientific notation. However, some of them are ambiguous, so answer “ambiguous” if so.

152 340 1000 9999 43210 0.0300 0.00213 31.0 31.00 31

 2.5 The expressions 0.123, 1.23, and 123 are numbers with **three** significant figures. Similarly, 1234 and 1.234×10^{-3} are numbers with **four** significant figures. How about the following expressions?

- | | | | |
|------------|--------------|--------------------------|----------------------------|
| (1) 3.14 | (2) 0.11 | (3) 1.1×10^{-2} | (4) 1.1×10 |
| (5) 1.0008 | (6) 1.0020 | (7) 1.0000 | (8) 4.00×10^{-10} |
| (9) 0.0008 | (10) 0.00080 | (11) 12345 | (12) 67890 |

2.3 Calculation with Significant figures

We need to do calculations of numbers with uncertainties, such as $1.23 + 4.56$ or $1.23/4.56$, but how? Here we discuss a simple method for

- addition ($x + y$) and subtraction ($x - y$)
- multiplication ($x \times y$) and division ($x \div y$ or x/y)

Other calculations, such as $\sqrt{x}$, e^x or $\sin(x)$, need the professional method given in  TODO  chap.

Multiplication and Division

As a rule, if x has m significant figures and y has n significant figures, then $x \times m$ or $x \div y$ should be rounded to have $\min(m, n)$ significant figures. ...But the rule is tough to understand. Probably you should see the example, and learn through practice.

Example 2.2

Calculate the following expressions, using calculators.

1. 8.912×2.4

2. $5.0 \div 3.14$

3. 2.01×49.8

4. 1.210^3

Solution

Here we use a shorthand notation “3-SF” to mean “three significant figures”.

(1) Since 8.912 has 4-SF and 2.4 has 2-SF, we round the result to (the smaller) 2-SF:

$$8.912 \times 2.4 = 21.3888 \rightarrow \underline{21}.$$

(2) Since 5.0 has 2-SF and 3.14 has 3-SF, we round the result to 2-SF:

$$5.0 \div 3.14 = 1.592... \rightarrow \underline{1.6}.$$

(3) Both 2.01 and 49.8 have 3-SF, so we keep 3-SF, but don't forget to avoid ambiguity!

$$2.01 \times 49.8 = 100.098 \rightarrow 100 \rightarrow \underline{1.00 \times 10^2}.$$

(4) $1.210^3 = 1.210 \times 1.210 \times 1.210$, so we round the result to 4-SF:

$$1.210^3 = 1.771561 \rightarrow \underline{1.772}.$$

If you want, we can justify this treatment by carrying out a hand calculation of the long multiplication.

$$\begin{array}{r} 1.20 \\ \times 1.31 \\ \hline 120 \\ 360 \\ 120 \\ \hline 15720 \end{array}$$

$$\begin{array}{r} 8.912 \\ \times 2.4 \\ \hline 35648 \\ 17824 \\ \hline 213888 \end{array}$$

$$\begin{array}{r} 2.01 \\ \times 49.8 \\ \hline 1608 \\ 1809 \\ 804 \\ \hline 100.098 \end{array}$$

Observe which digits are uncertain, and how they “pollute” the results in each step. In 1.20×1.31 , the last **1** pollutes the part of **1** + 6 and we get an uncertain number **7**. So, the result is **1.57**.

Addition and subtraction

Different rules are applied for addition and subtraction. You can understand the rules easily if you do long addition and observe which digits are polluted.

Example 2.3

Calculate the following expressions.

1. $12.3 + 4.56$

2. $123 + 4.56$

3. $0.50 - 0.032$

4. $1.20 \times 10^3 - 27$

Solution

$$\begin{array}{r} 12.3 \\ + 4.56 \\ \hline 16.86 \end{array}$$

$$\begin{array}{r} 123 \\ + 4.56 \\ \hline 127.56 \end{array}$$

$$\begin{array}{r} 0.50 \\ - 0.032 \\ \hline 0.468 \end{array}$$

$$\begin{array}{r} 1200 \\ - 27 \\ \hline 1173 \end{array}$$

and we round the results. So, the answers are **16.9**, **128**, **0.47**, and **1.17×10^3** .

Problems

 **2.6** Calculate the following, taking care of significant figures. You may use calculators.

- | | | | |
|--|---|------------------------|-----------------------------|
| (1) 1.23×4.5 | (2) 1.50×2.0 | (3) 2.00×5.00 | (4) 4.0×2.5 |
| (5) $1.23 + 4.5$ | (6) $1.50 + 2.0$ | (7) $2.00 + 8.00$ | (8) $100.0 + 0.123$ |
| (9) $1.23 \div 4.5$ | (10) $6.0 \div 3.00$ | (11) $9.00 \div 4.5$ | (12) $1.50 \div 3.0$ |
| (13) $1.23 - 4.5$ | (14) $3.10 - 0.10$ | (15) $3.10 - 0.1$ | (16) $3.10 - 3.0$ |
| (17) $131 + 69$ | (18) $131 + 6.9$ | (19) $131 + 0.69$ | (20) $1.3 \times 10^2 + 69$ |
| (21) $(1.2 \times 10^5) \times 9.4$ | (22) $1.2 \div (5.2 \times 10^5)$ | | |
| (23) $(3.33 \times 10^5) \times (6.3 \times 10^4)$ | (24) $(3.33 \times 10^5) \times (6.3 \times 10^{-4})$ | | |
| (25) $(3.33 \times 10^5) \div (6.3 \times 10^4)$ | (26) $(3.33 \times 10^5) \div (6.3 \times 10^{-4})$ | | |
| (27) $1.00 \times 10^4 + 2.5 \times 10^3$ | (28) $3.00 \times 10^3 - 4.50 \times 10^2$ | | |
| (29) $2.99 \times 10^2 + 0.814$ | (30) $1.23 \times 10^{-4} + 4.5 \times 10^{-6}$ | | |

****2.7** Calculate the following, taking care of significant figures. You may use calculators.

- | | | | |
|-------------------------------------|--------------------------------|---|-----------------|
| (1) $1.50 \times 2.000 \times 3.50$ | (2) $8.00 \div 4.00 \div 0.25$ | (3) 5.0^3 | (4) 2.10^3 |
| (5) $1.0 + 2.0 + 3.0 \times 4.0$ | (6) $17 - 8.0 - 3.0$ | (7) $3.0^3 + 1.2$ | (8) $5.0^3 - 5$ |
| (9) $(1.2 + 3.45) \times 2.1$ | (10) $(8.0 - 1.25) \div 2.0$ | (11) $1.2 \times 3.4 + 5.6$ | |
| (12) $18.0 - 2.5 \times 1.2$ | (13) $1.20 \times 3.40 + 5.60$ | (14) $(5.0 \times 10^2 + 3.4) \times 1.2$ | |

 **2.8** Calculate the following with calculators, taking care of significant figures and units.

- | | |
|---|--|
| (1) $3.1 \text{ m/s} \times 1.0 \text{ hour}$ | (2) $3.1 \text{ m} \div 25 \text{ cm}$ |
| (3) $1.5 \text{ kg} \times 9.8 \text{ m/s}^2$ | (4) $1.50 \text{ km} + 195 \text{ m}$ |
| (5) $(5.2 \text{ kg/m}^3) \times (4.0 \text{ }\mu\text{m})^3$ | (6) $(5.2 \text{ kg/m}^3) \times (4.0 \text{ }\mu\text{m}^3)$ |
| (7) $1.2 \text{ }\mu\text{m} \div 2.00 \text{ nm}$ | (8) $(1.7 \times 10^{-27} \text{ kg}) \times (3.00 \times 10^8 \text{ m/s})^2$ |
| (9) $2.000 \text{ kg} \times (3.0 \text{ m/s}^2)$ | (10) $2.000 \text{ kg} \times (3.0 \text{ m/s})^2$ |
| (11) $(1.60 \times 10^{-19} \text{ C}) \div 2.0 \text{ }\mu\text{s}$ | (12) $(1.60 \times 10^{-19} \text{ C}) \times 5.0 \text{ kV}$ |
| (13) $(6.63 \times 10^{-34} \text{ J} \cdot \text{s}) \times (3.0 \times 10^{12} \text{ Hz})$ | (14) $1.50 \text{ MJ} \div 20.0 \text{ m}$ |

Advanced note: Uncertainties can often be determined subjectively, but central values are also without specific definitions; it is often the averaged value of the measurements, but one may assume some probabilistic distribution and use its median, mean, or mode as the central value.

Chapter 3

Logic

Solve [Quiz 2.3](#) (on page 19) again.

- (1) If $mr\omega^2 = 10.0 \text{ kg}\cdot\text{m/s}^2$, $m = 5.0 \text{ kg}$, and $r = 1.0 \text{ m}$, what is ω ?
- (2) If $v_0t + \frac{1}{2}at^2 = 5.0 \text{ m}$, where $a = -4.0 \text{ m/s}^2$ and $v_0 = 7.0 \text{ m/s}$, what is t ?
- (3) At time $t = 0$, a particle is at $(x, y) = (2.0 \text{ m}, 0)$. It moves with a constant velocity $(v_x, v_y) = (3.0, 4.0) \text{ m/s}$. What is its position at $t = 1.0 \text{ s}$?

The answers are

- (1) $\omega = \pm 1.4 \text{ s}^{-1}$ (2) $t = 1.0 \text{ s}, 2.5 \text{ s}$ (3) $x = 5.0 \text{ m}, y = 4.0 \text{ m}$

but **what do they mean?**

Quiz

Q1. Guess the correct meaning of each of above expressions.

- | | | |
|--|------|---|
| (1) ω is both $+1.4 \text{ s}^{-1}$ and -1.4 s^{-1} | v.s. | ω is either $+1.4 \text{ s}^{-1}$ or -1.4 s^{-1} |
| (2) t is both 1.0 s and 2.5 s | v.s. | t is either 1.0 s or 2.5 s |
| (3) x is 5.0 m and y is 4.0 m | v.s. | Either x is 5.0 m , or y is 4.0 m . |

Elementary educations do not discuss these differences because kids do not know **logical thinking**. Now, you are a grown-up university student. You need to think about it.

3.1 Basic Logics

Since elementary school, you have written many equalities and inequalities, such as

$$3 + 5 = 8, \quad 6 + 2 = 10, \quad 1 + 3 > -3, \quad 1 + 3 \neq 0, \quad 3^3 = 9, \quad 1 > 2, \quad \text{and} \quad \sin(\pi) \neq 0.$$

Quiz

Q2. The above statements are either **true** (T) or **false** (F). For each of them, state whether it is true or false.

Q3. Confirm that the following statements are all true:

- | | |
|-------------------------------|-------------------------------|
| (1) $3 + 5 = 8$ is true. | (2) $6 + 2 = 10$ is false. |
| (3) $3 \neq 3$ is false. | (4) False is false. |
| (5) "False is false" is true. | (6) "False is true" is false. |

We can only write true things, such as " $3 + 5 = 8$ ", " $3^2 = 9$ is true", or " $6 + 2 = 10$ is false". In other words, **when you write something, you must confirm that it is true**. Sometimes, you are asked to confirm a statement, or **prove** a statement. The process is called **proof** of the statement.

Numbers can be manipulated by operators such as $+$, $\div$. Similarly, true (T) and false (F) can be manipulated by **and**, **or**, and **not** operators.

and (conjunction) “A and B” ($A \wedge B$) is true if both A and B are true, and false otherwise.

or (disjunction) “A or B” ($A \vee B$) is true if at least one of A and B is true. False if both are false.

not (negation) “not A” ($\neg A$) is true if A is false, and false if A is true.

We can write these property as equations:

$$(T \wedge T) = T, \quad (T \wedge F) = (F \wedge T) = (F \wedge F) = F,$$

$$(F \vee F) = F, \quad (T \vee T) = (T \vee F) = (F \vee T) = T,$$

$$(\neg T) = F, \quad (\neg F) = T,$$

A	B	A and B	A or B	not A
T	T	T	T	F
T	F	F	T	F
F	T	F	T	T
F	F	F	F	T

but the **truth table**, shown to the right, is more useful.

Remark: Note that “ $1 + 1 = 2$ or $2 + 2 = 4$ ” is true, which says the word “or” is a bit different from everyday English. In daily conversation, “*Sho will eat ramen or sushi tonight*” usually means “but not both”. However, in mathematics, if Sho eats both ramen and sushi for a dinner, the statement “*Sho had sushi or ramen tonight*” is true.

Quiz

Q4. State whether it is true or false for the following statements.

(1) $1 + 2 = 3$ and $3 + 4 < 5$.

(2) $10 \div 3 > 0$ or $10 \div 2 > 0$.

(3) $\neg(3 > 5)$.

(4) $(6 + 3 > 0) \wedge \neg(5 - 2 > 0)$.

Fail safe note: Don't be confused by daily English.

“ $x = 1$ and $x = -1$ are the two solutions of $x^2 = 1$.” (correct daily English)

“ $x^2 = 0$ if and only if $x = 1$ or $x = -1$.” (correct mathematical English)

are both correct, but if you say “ $x = 1$ and $x = -1$ ”, then it means an impossible equality $x = 1 = -1$.

3.2 Implication $\Rightarrow$

We can discuss the following statements: are they true or false?

Example 3.1

- If $x > 5$, then $x > 1$ This is true. (Number larger than 5 are larger than 1.)
- If $x < 9$, then $x < 2$ This is false, because we have a counterexample $x = 5$.
- If $x^2 < 1$, then $x < 1$ This is true.
- If $x^2 = 1$, then $x = -1$ This is false, because we have a counterexample $x = 1$.

In general, it is easy to claim that a statement is false, i.e., to *refute* or *disprove* a statement. You just have to find one **counterexample**. Meanwhile, it is more difficult to claim that a statement is true. As mentioned above, you need to write a proof.

Now, let's try to find some counterexamples.

Quiz

Q5. The following statements are false. Find a counter example for each.

- (1) If $x^2 > 0$, then $x > 0$. (2) If $x^2 - x = 0$, then $x = 0$.
 (3) If $x^2 = y^2$, then $x = y$. (4) If $x + y < 1$, then $x < 1$ and $y < 1$.

We use the symbol “ $\Rightarrow$ ” to express “if ... then ...” statement.

$A \Rightarrow B$ means “If A is true, then B is true.” or equivalently, “ A implies B .”

and this is called **implication**. For example, the statements in the above quiz can be written as

$$x^2 > 0 \Rightarrow x > 0, \quad x^2 - x = 0 \Rightarrow x = 0, \quad x^2 = y^2 \Rightarrow x = y, \quad x + y < 1 \Rightarrow x < 1 \wedge y < 1.$$

Quiz

Q6. Similarly rewrite the statements in Example 3.1 with using the symbol “ $\Rightarrow$ ”.

Advanced note: In formal logic, $A \Rightarrow B$ is defined by $(\neg A) \vee B$, which means “if A is true, B must also be true; but if A is false, we do not care about B ”. So, if A is always false, $A \Rightarrow B$ is always true, no matter what B is. For example, $(x \in \mathbb{R} \wedge x^2 < -1) \Rightarrow x = 999$ is a true statement. Through this definition, it is clear that $A \Rightarrow B$ is equivalent to $\neg B \Rightarrow \neg A$, which we will see below.

3.3 Equivalent Statements—What does “solve” mean?

Since elementary school, you have solved many equations, but what does “solving an equation” mean?

Quiz

Q7. (1) Choose the true statements.

- (a) $2x - 1 = 0 \Rightarrow x = 1/2$. (b) $x^2 = 4 \Rightarrow x = 2$.
 (c) $2x - 1 = 0 \Rightarrow x > -10$. (d) $x^2 = 4 \Rightarrow x = 0$.

(2) Do you think these statements can be considered as “solving an equation”?

(b) and (d) are false statements (find counterexamples!), so we cannot write them. (a) and (c) are true statements, and (a) looks “solving an equation”, but we do not consider (c) is “solving an equation”. The difference between (a) and (c) is that

$$2x - 1 = 0 \Leftarrow x = 1/2 \text{ is true, but } 2x - 1 = 0 \Leftarrow x > -10 \text{ is false.}$$

So, for (a), both $\Rightarrow$ and $\Leftarrow$ are correct: we use $\Leftrightarrow$ to express such cases.

Definition 3.2 (Equivalent)

If both $A \Rightarrow B$ and $A \Leftarrow B$ are true, we call “ A and B are **equivalent**”, and write $A \Leftrightarrow B$.

$A \Rightarrow B$ is read by “ B if A ” in English. Similarly, $A \Leftrightarrow B$ is read by “ B if and only if A ” or “ B **iff** A ”. (namely, **if-and-only-if**).

Now you need to do a bit of practice. It is actually tough, unfortunately, but this drill is important for logical thinking.

Problems

 **3.1** Fill in the blanks with $\Rightarrow$, $\Leftarrow$, or $\Leftrightarrow$, where x is a real number.

- | | | |
|---|--|--|
| (1) $x = 2$ <input type="text"/> $x > 0$ | (2) $x > 1$ <input type="text"/> $x > 0$ | (3) $x = 1$ <input type="text"/> $x > 0$ |
| (4) $x = -3$ <input type="text"/> $x^2 = 9$ | (5) $x = 0$ <input type="text"/> $x^2 = 0$ | (6) $x < 0$ <input type="text"/> $x^2 > 0$ |
| (7) $x = 3$ <input type="text"/> $x \neq 5$ | (8) $3x = 3$ <input type="text"/> $x = 1$ | (9) $x^3 = x^2$ <input type="text"/> $x = 1$ |

If the statement “ $A \Rightarrow B$ ” is true, then

- B is called a **necessary condition** for A , because B is necessary for A ; if not B , then not A .
- A is called a **sufficient condition** for B , because if A is true, B is “sufficiently” true.

So, if A and B are equivalent, B (resp. A) is called *necessary-and-sufficient condition* for A (resp. B). For example, because $x = 1 \Rightarrow x^2 = 1$,

- $x = 1$ is sufficient to satisfy $x^2 = 1$. ($x = 1$ is a sufficient condition for $x^2 = 1$.)
- $x^2 = 1$ is necessary to satisfy $x = 1$. ($x^2 = 1$ is a necessary condition for $x = 1$.)

Quiz

Q8. Fill in the blanks with “a sufficient condition” or “a necessary condition”.

- | | |
|--|---|
| (1) $x = -1$ is <input type="text"/> for $x^2 = 1$. | (2) $x^2 = 4$ is <input type="text"/> for $x = 2$. |
| (3) $x > 1$ is <input type="text"/> for $x > 0$. | (4) $x^2 > 0$ is <input type="text"/> for $x < 0$. |
| (5) $x < 0$ is <input type="text"/> for $x^2 > 0$. | (6) $x^2 > 0$ is <input type="text"/> for $x > 0$. |



“To solve an equation” means “to find an *equivalent* equation in the form of $x = \boxed{}$ ”. Since $2x - 1 = 0 \Leftrightarrow x = 1/2$ and $x^2 = 4 \Leftrightarrow x = \pm 2$, we say $x = 1/2$ and $x = \pm 2$ are the solutions of the equations, respectively.

When you solve an equation, you have to check that the solution is **necessary and sufficient**.

- If you don’t check necessity, you may have an “incomplete solution”.
- If you don’t check sufficiency, you may have an “extraneous solution”.

Quiz

- Q9.** (1) Solve $\sqrt{x+2} = x$. You might reach an *extraneous solution*, which you need to *exclude* it.
- (2) Solve $\sqrt{x^2} = 4$. You might reach an *incomplete solution*, where you need to find more solutions. (Hint: $\sqrt{x^2} = x$ is a false statement.)

Before discussing advanced topics on logics, you should do some drills.

Problems

 **3.2** State whether each statement is true (T) or false (F). For example, $2 > 5$ is “F”.

- | | | |
|----------------------------------|------------------------------------|--------------------------------------|
| (1) $(3 > 1) \wedge (2 < 5)$ | (2) $3 < 1 \vee 2 > 5$ | (3) $\neg(3 > 1)$ |
| (4) $\neg(2 > 5)$ | (5) $(\neg(3 < 1)) \wedge (2 < 5)$ | (6) $\neg(3 < 1 \wedge 2 < 5)$ |
| (7) $(4 = 4) \wedge \neg(2 > 5)$ | (8) $(3 < 1) \vee \neg(3 = 1)$ | (9) $(\neg(3 < 1)) \vee \neg(3 > 1)$ |

 **3.3** State whether each statement is true or false. If false, give a counter example.

- | | |
|--|--|
| (1) If $x = 2$, then $x^2 = 4$. | (2) If $x^2 = 4$, then $x = 2$. |
| (3) $x^2 = 0 \implies x = 0$ | (4) $x > 2 \implies x > 0$ |
| (5) $x > 0 \implies x > 2$ | (6) $x = 3 \implies x = 3$ |
| (7) $(x = 1) \vee (x = -1) \implies x^2 = 1$ | (8) $x^2 = 1 \implies (x = 1) \vee (x = -1)$ |

★3.4 Solve each equation, assuming x is a real number.

- | | | | |
|----------------------|----------------------|-----------------|------------------|
| (1) $\sqrt{x} = 3$ | (2) $\sqrt{x-1} = 2$ | (3) $ x = 5$ | (4) $ x = -3 $ |
| (5) $\sqrt{x^2} = 3$ | (6) $ x^2 = 5$ | (7) $ x-2 = 3$ | (8) $ 2x+1 = 7$ |

****3.5** Solve each equation, assuming x is a real number.

- | | | |
|----------------------------|----------------------------|----------------------------|
| (1) $\sqrt{x-3} = 2$ | (2) $\sqrt{x-3} = x-3$ | (3) $\sqrt{x-3} = 3-x$ |
| (4) $\sqrt{(x-3)^2} = 2$ | (5) $\sqrt{(x-3)^2} = x-1$ | (6) $\sqrt{(x-3)^2} = x-3$ |
| (7) $\sqrt{(x-3)^2} = 3-x$ | (8) $\sin x = 1/2$ | (9) $\tan x = 0$ |

★3.6 Let x and y be real numbers. Fill in the blanks with $\implies$, $\impliedby$, or $\iff$. If your answer is $\implies$ or $\impliedby$, explain the reason by giving relevant counter examples.

- | | |
|---|--|
| (1) $x = 2$ ☐ $x^2 = 4$ | (2) $x^2 = 4$ ☐ $x = 2 \vee x = -2$ |
| (3) $x > 0$ ☐ $x^2 > 0$ | (4) $ x = 2$ ☐ $(x = 2 \vee x = -2)$ |
| (5) $x + 1 = 0$ ☐ $x = -1$ | (6) $x^2 = x$ ☐ $x(x-1) = 0$ ☐ $x = 0$ |
| (7) $\sqrt{x} = 123$ ☐ $x = 123^2$ | (8) $\sqrt{x^2} = 0$ ☐ $x = 0$ |
| (9) $\sqrt{x^2} = 3$ ☐ $x = 3$ | (10) $\sqrt{x^2} = 3 \wedge x > 0$ ☐ $x = 3$ |
| (11) $\sqrt{x^2} = \sqrt{y^2}$ ☐ $x = y$ | (12) $x + y > 0$ ☐ $x > 0 \wedge y > 0$ |
| (13) $xy = 0$ ☐ $x = y = 0$ | (14) $x^2 + y^2 = 0$ ☐ $x = y = 0$ |

****3.7** Let a be a real constant and x be a real number. For (1)–(4), fill in the blanks with $\implies$, $\impliedby$, or $\iff$. If your answer is $\implies$ or $\impliedby$, explain the reason by counter examples.

(1) $\sqrt{x} = 4$ $x = 16$

(2) $\sqrt{x^2} = 4$ $x = 4$

(3) $\sqrt{x} = a^2$ $x = a^4$

(4) $\sqrt{x^2} = a^2$ $x = a^2$

(5) Solve the equation $\sqrt{x^2} = a$, noting that a can be negative.

★3.8 The following “solutions” have logical errors. Identify errors and find the right answer.

(1) A student solves $x^2 = 9$ and writes $x = 3$.

(2) He solves $x^2 - 3x = 0$ by dividing both sides by x , obtaining $x - 3 = 0$, so $x = 3$.

(3) He solves $x^6 = x^4$ by dividing both sides by x^4 , obtaining $x^2 = 1$, so $x = \pm 1$.

Next, fill in the blanks with $\implies$, $\impliedby$, or $\iff$.

(4) $x^2 = 9$ $x = 3$ (5) $x^2 - 3x = 0$ $x - 3 = 0$ $x = 3$

(6) $x^6 = x^4$ $x^2 = 1$ $x = \pm 1$

****3.9** (1) Recall that $A \Rightarrow B$ is defined by $(\neg A) \vee B$. Also, recall that $A \Leftrightarrow B$ is defined by $A \Rightarrow B \wedge A \Leftarrow B$. Write a truth table (see page 26) for $A \Rightarrow B$, $A \Leftarrow B$, and $A \Leftrightarrow B$.

(2) Explain the reason we can understand $A \Leftrightarrow B$ as $A = B$.

(3) Write a truth table for the following expressions:

$$A \wedge B, \quad \neg(A \wedge B), \quad (\neg A) \vee (\neg B), \quad \neg(A \vee B), \quad (\neg A) \wedge (\neg B)$$

Explain that this truth table is considered as a *proof* of **de Morgan's theorem**

$$\neg(A \vee B) = (\neg A) \wedge (\neg B), \quad \neg(A \wedge B) = (\neg A) \vee (\neg B). \quad (3.1)$$

(4) Prove the following, which we will discuss in the next section.

$$(A \Rightarrow B) = ((\neg B) \Rightarrow (\neg A)), \quad (A \Rightarrow B) = \neg(A \wedge \neg B). \quad (3.2)$$

****3.10** This lecture does not cover **quantifiers** “for-all $\forall$ ” and “exists $\exists$ ”. Learn them by yourselves. In particular, prove the following:

(1) $\exists x, P(x) \iff \neg(\forall x, \neg P(x))$

(2) $\neg(\exists x, P(x)) \iff \forall x, \neg P(x)$

(3) $\forall x, P(x) \iff \neg(\exists x, \neg P(x))$

(4) $\neg(\forall x, P(x)) \iff \exists x, \neg P(x)$

****3.11** For each statement, write its negation, such as “not ($x \geq 0$ for all real x)”, in a natural form. Then, state whether the original statement and the negated one are true or false.

(1) $x^2 \geq 0$ for all real x .

(2) There exists a real x such that $x^2 = -1$.

(3) For all real x , $x > 0 \Rightarrow x^2 > 0$.

(4) $|x| = x$ for all real x .

3.4 A few more notes about logic and proof

There is a useful theorem for “not” operator, called **de Morgan’s theorem**.

Theorem 3.3 (De Morgan's theorem)

1. “not (A and B)” is equivalent to “(not A) or (not B)”, i.e., $\neg(A \wedge B) = (\neg A) \vee (\neg B)$.
2. “not (A or B)” is equivalent to “(not A) and (not B)”, i.e., $\neg(A \vee B) = (\neg A) \wedge (\neg B)$.

The proof is given as [Problem 3.9](#).

Quiz

Q10. Simplify the following statements by using de Morgan’s theorem.

- | | |
|--------------------------------------|-------------------------------|
| (1) not ((not A) or (not B)) | (2) not ((not A) and (not B)) |
| (3) $\neg((\neg A) \wedge (\neg B))$ | (4) $\neg(A \vee \neg B)$ |

Q11. For each of the following, write its negation (“not A”).

- | | | |
|--------------------------|---------------------------------|---------------------------------|
| (1) $x > 0$ or $x < 1$. | (2) $x \neq 4$ and $x \neq 5$. | (3) $0 < x < 1$. |
| (4) $x > 0$ or $y > 0$. | (5) $x = 0$ and $y = 0$. | (6) $x \neq 0$ and $y \neq 0$. |

In general, $A \Rightarrow B$ and $B \Rightarrow A$ are different. However, if $A \Rightarrow B$, then $\neg B \Rightarrow \neg A$ (not $B \Rightarrow$ not A) is always correct. It is called the **contrapositive** of $A \Rightarrow B$. Let’s see an example.

Example 3.4

If $0 < x < 2$, then x^2 is always less than 4. We can write it by $(0 < x < 2) \Rightarrow (x^2 < 4)$. Let A be the statement “ $0 < x < 2$ ” and B be the statement “ $x^2 < 4$ ”. “Not A ” is “ $x \leq 0$ or $x \geq 2$ ”. Meanwhile, “not B ” is “ $x^2 \geq 4$ ”.

- Its **converse** ($A \Leftarrow B$) is $(x^2 < 4) \Rightarrow (0 < x < 2)$, which is false (counterexample: $x = -1$).
- Its **inverse** (not $A \Rightarrow$ not B) is $(x \leq 0 \text{ or } x \geq 2) \Rightarrow (x^2 \geq 4)$, which is false.
- Its **contrapositive** (not $A \Leftarrow$ not B) is $(x^2 \geq 4) \Rightarrow (x \leq 0 \text{ or } x \geq 2)$, which is true.

Because the contrapositive of a true statement is always true, we can prove a statement by proving its contrapositive. This method is called **proof by contrapositive**.

Quiz

Q12. For each of the following statements, write its converse, inverse, and contrapositive. Then state whether each of them is true or false.

- | | |
|-----------------------------------|--------------------------------------|
| (1) If $x > 3$, then $x > 0$. | (2) If $x = 2$, then $x^2 = 4$. |
| (3) If $ x = 0$, then $x = 0$. | (4) If $x^2 = 4$, then $x \neq 3$. |

Q13. Prove the following statements by *proof by contrapositive*. Namely, first write the contrapositive, then prove it.

- (1) If $x^2 > 1$, then $|x| > 1$. (2) If $x^3 + x^2 + x < 0$, then $x < 0$.
 (3) If $xy \neq 0$, then $x \neq 0$. (4) If n^2 is not integer, then n is not integer.

Another powerful method is **proof by contradiction** (deductio ad absurdum). Imagine you prove “if A , then B ”. You assume “ A ” and “not B ” are both true, and derive a contradiction. If you reach a contradiction, it means “ A and not B ” is false, i.e., “not (A and not B)” is true, which means $A \Rightarrow B$.

Quiz

Q14. Prove the following statements by *proof by contradiction*.

- (1) If x^2 is an even integer, then x cannot be an odd integer.
 (2) If a , b , and c are integers and abc is even, at least one of a , b , or c is even.

Remark: The validity of “proof by contrapositive” and “proof by contradiction” was already discussed on page 30, Eq. (3.2).

Advanced note: You learned another useful method for proof, **mathematical induction**, in high school. In university, you may learn more advanced methods, such as **infinite descent** and **transfinite induction**.

3.5 Facts, Definitions, Assumptions, and Conclusions

One of the main reasons students find university physics difficult is that they treat all statements in the same way. In physics, every statement has its own role:

- experimental fact
- definition (in mathematics: definition)
- assumption / approximation (in mathematics: axiom or assumption)
- derived statements (in mathematics: theorem, corollary, or proposition)

You should always know which role each statement plays.

Remark: Mathematics does not have experimental facts because it is not based on experiments; it is purely based on definitions.

Example 3.5 (Types of statements in a calculation)

A particle moves as $x(t) = kt^2 + x_0$, where k and x_0 are constants. Find the averaged velocity between $t = 0$ and $t = t_0$.

Solution

We should first clarify assumptions and definitions:

- Assumptions (explicitly written): $x(t) = kt^2 + x_0$. k and x_0 are constants.
- Assumptions (implicit): k , x_0 , and t_0 are real numbers, and t means time.

- Definition: averaged velocity is *defined* by $v_{\text{avg}} := \frac{x(t_0) - x(0)}{t_0 - 0}$.

Now, we can calculate $\frac{x(t_0) - x(0)}{t_0 - 0} = \frac{(kt_0^2 + x_0) - x_0}{t_0} = kt_0$ to give a derived statement:

- Derived statement (conclusion): the averaged velocity between $t = 0$ and t_0 is $v_{\text{avg}} = kt_0$.

Quiz

Q15. Carry out a similar discussion for “find the instantaneous velocity at $t = t_1$ ”.

Consider an **experimental fact**, such as “gravity is always attractive”. We have to accept it. We need to remember it and understand what it says. “Protons have charge $+|e|$ and electrons have charge $-|e|$ ” is another experimental fact, but we can understand it as the definition of the elementary charge $|e|$.

Definitions often appear as mathematical equations, such as “instantaneous velocity is defined by $v(t) := dx/dt$ ”. Another example is “uniform circular motion is a motion with constant speed along a circle”; here, the word “uniform circular motion” is defined.

Meanwhile, **assumption** is what we can impose freely. For example, you can decide whether you ignore air resistance or not. However, every conclusion is valid only under its assumptions. If you impose bad assumptions, your conclusion will be meaningless. For example, if you want to analyze the free fall of a sheet of paper but you ignore air resistance, your result should be invalid and meaningless. You must be very careful when you make assumptions.

Remark: We often use “ $A := B$ ” to say “define A by B ”. For example, $v(t) := dx/dt$ is the definition of instantaneous velocity, and $T := 1/f$ is the definition of period. Meanwhile, “ $A = B$ ” just means “ A is equal to B ”.

Some physicists use $A \equiv B$ to say “ A is defined by B ”, but Sho does not recommend it because mathematicians use the symbol $\equiv$ for other purposes.

Problems

★3.12 Prove the following statements by contrapositive or contradiction.

- (1) $x^2 + y^2 = 0 \implies x = 0$.
- (2) $ab = 0 \implies a = 0$ or $b = 0$.
- (3) $ab = 0 \iff a = 0$ or $b = 0$.

★3.13 With de Morgan’s theorem, rewrite the following statements in a form without “not”.

- (1) not ($x > 0$ and $y > 0$)
- (2) not ($x = 0$ or $y = 0$)
- (3) not (if $x > 1$, then $x^2 > 1$)
- (4) not ($x > 0$ iff $x^2 > 0$)

★3.14 Find the logical error: Let $a = b = 1$. Then $a^2 = ab$, so $a^2 - b^2 = ab - b^2$. So, $(a - b)(a + b) = b(a - b)$, which gives $a + b = b$, so $2 = 1$.

***3.15 For each of the following statements, write its converse, inverse, and contrapositive. State T or F for the original statement, its converse, inverse, and contrapositive.

- (1) If $x^2 - 5x + 6 = 0$, then $x = 2$ or $x = 3$.
- (2) If $x > 0$ and $y > 0$, then $x + y > 0$.
- (3) If x and y are both even integers, then xy is divisible by 4.

***3.16 A student claims: "I proved $\neg B \Rightarrow \neg A$, so I have proved $B \Rightarrow A$." Is this correct? Explain, and state what the student has actually proved.

***3.17 The following two statements look similar but are logically different. For each, state whether it is T or F and explain.

- (1) For all real x : if $x > 0$, then $x^2 > 0$.
- (2) For all real x : if $x^2 > 0$, then $x > 0$.

***3.18 True or false? Give a reason.

- (1) The contrapositive of a false statement is also false.
- (2) If $A \Rightarrow B$ is true and $B \Rightarrow A$ is false, then A and B are not equivalent.
- (3) If A is false, then $A \Rightarrow B$ is always true, for any B .

*3.19 (1) What is a **proposition**?

(2) Prove the following statements on propositions P , Q , and R .

- (a) $(P \Rightarrow Q \wedge Q \Rightarrow R) \implies P \Rightarrow R$. (**syllogism**)
- (b) $(P \Rightarrow Q) \wedge (P \Rightarrow R) \iff P \Rightarrow (Q \wedge R)$.

Chapter 4

Power, Exponential, and Logarithm

4.1 Powers

Let's begin with a review: can you recall all the rules for powers...?

Quiz

Q1. Calculate without a calculator.

(1) 2^4 , 3^5 , 4^2 , 10^3 , 1^{25} , 2^{10}

(2) 5^0 , 3^0 , 2^0 , 1^0

(3) 2^{-1} , 12^{-1} , 3^{-3} , 2^{-4} , 5^{-2} , 2^{-10}

(4) $\left(\frac{1}{2}\right)^{-1}$, $\left(\frac{1}{3}\right)^{-2}$, $\left(\frac{1}{12}\right)^{-1}$, $\left(\frac{1}{5}\right)^{-2}$, $\left(\frac{1}{2}\right)^{-10}$, $\left(\frac{1}{3}\right)^0$, $\left(\frac{1}{10}\right)^0$

(5) $\left(\frac{1}{2}\right)^4$, $\left(\frac{2}{3}\right)^3$, $\left(\frac{5}{2}\right)^2$, $\left(\frac{2}{5}\right)^{-2}$, $\left(\frac{7}{2}\right)^2$, $\left(\frac{7}{2}\right)^{-2}$, $\left(\frac{1}{3}\right)^2$, $\left(\frac{1}{3}\right)^{-2}$.

(6) $4^{1/2}$, $81^{1/2}$, $3^{1/2}$, $27^{1/3}$, $\left(\frac{1}{4}\right)^{1/2}$, $\left(\frac{1}{7}\right)^{1/2}$, $\left(\frac{8}{9}\right)^{1/2}$, $\left(\frac{15}{3}\right)^{1/2}$

(7) $4^{-1/2}$, $81^{-1/2}$, $3^{3/2}$, $27^{2/3}$, $27^{-2/3}$, $\left(\frac{1}{4}\right)^{3/2}$, $\left(\frac{1}{7}\right)^{-1/2}$, $\left(\frac{1}{2}\right)^{-5/2}$

Q2. Fill in the blanks, where $A > 0$, $B > 0$, and x, y are real numbers.

(1) $A^3 \cdot A^3 = A^{\square}$ (2) $(A^3)^3 = A^{\square}$ (3) $(A \cdot B)^4 = \square \cdot \square$

(4) $A^x \cdot A^y = A^{\square}$ (5) $(A^x)^y = A^{\square}$ (6) $(A \cdot B)^x = A^x \cdot \square$

(7) $A^{\square} = \frac{1}{A^x}$ (8) $A^x = \frac{1}{\square}$ (9) $\frac{A^x}{A^y} = A^{\square}$

(10) $A^0 = \square$ (11) $A^{\square} = A$ (12) $A^{\square} = \sqrt{A}$

(13) $A^{\square} = \sqrt[3]{A}$ (14) $\frac{A^x}{B^x} = \square$ (15) $\left(\frac{A}{B}\right)^{-1} = \square$

Q3. With $A > 0$, $B > 0$, $x \in \mathbb{R}$, and $y \in \mathbb{R}$, the following statements are all false. Find a counterexample for each.

(1) $A^x + A^y = A^{x+y}$ (2) $(A + B)^x = A^x + B^x$ (3) $(A \cdot B)^x = A \cdot B^x$

(4) $(A^x)^y = A^{(x^y)}$ (5) $(A^x)^y = A^{x+y}$

Do not worry if you cannot recall all the rules! Learn this section and come back here, and then you will be able to answer all the questions!

Let's review the definitions of **power**, a^x . Here, a is called the **base** and x is called the **exponent**.

Definition 4.1 (Power for positive base, real exponent)

For positive base $a > 0$ and real exponent $x \in \mathbb{R}$, we define the power a^x by the following steps.

First, consider a non-negative integer n . We define a^n and n -th root $\sqrt[n]{a}$, as

- $a^0 := 1$.
- If $n \in \mathbb{N}^+$, then $a^n := a \cdot a^{n-1}$.
- For $n \in \mathbb{N}^+$, we define $\sqrt[n]{a}$ as the **positive** solution X of the following equation: $X^n = a$. We write $\sqrt[2]{a}$ as $\sqrt{a}$.

If $a > 0$, positive solution always exists and is unique. So, $\sqrt[n]{a}$ is “well-defined” for $a > 0$. Then,

- If $x > 0$ and $x \in \mathbb{Q}$, we can write $x = p/q$ with $p, q \in \mathbb{N}^+$. Then we define

$$a^x = a^{p/q} := \left(\sqrt[q]{a}\right)^p = \left(a^{1/q}\right)^p$$

- If $x > 0$ and $x \notin \mathbb{Q}$, we can consider a sequence of rational numbers $x_k \in \mathbb{Q}$ such that $x_k \rightarrow x$. Then we define $a^x := \lim_{k \rightarrow \infty} a^{x_k}$.

Now we have defined a^x for all $x \geq 0$. Finally,

- If $x < 0$, we define $a^x := 1/(a^{|x|})$.

and then we have defined a^x for all $x \in \mathbb{R}$, if $a > 0$.

Example 4.2 (Integer-power of a positive number)

- $2^0 := 1$, because of the first item in the above definition.
- The second item says $2^3 := 2 \times 2^2$. Using it recursively, we get $2^2 := 2 \times 2^1$ and $2^1 = 2 \times 2^0 = 2 \times 1 = 2$. So, $2^3 = 2 \times 2 \times 2 = 8$.
- Similarly, $1.1^2 = 1.1 \times 1.1 = 1.21$ and $1.1^3 = 1.1 \times 1.21 = 1.331$.

Example 4.3 (n-th root of a positive number)

The third item says $\sqrt[3]{8}$ is the positive solution of $X^3 = 8$. So, $\sqrt[3]{8} = 2$. Similarly, $\sqrt[2]{1.21}$ is the positive solution of $X^2 = 1.21$, so $\sqrt{1.21} = 1.1$.

Example 4.4 (positive-power of a positive number)

To consider a^x for rational but non-integer x , we use the fourth item of the above definition.

- $27^{1/3} = \left(\sqrt[3]{27}\right)^1 = 3^1 = 3$. • $1.21^{0.5} = 1.21^{1/2} = \sqrt{1.21} = 1.1$.
- $27^{2/3} = \left(\sqrt[3]{27}\right)^2 = 3^2 = 9$. • $1.21^{1.5} = 1.21^{3/2} = \left(\sqrt{1.21}\right)^3 = 1.1^3 = 1.331$.

If x is not rational, we need to use the fifth item. For example, if $x = \sqrt{2} = 1.4142\dots$, we consider a sequence $1, 14/10, 141/100, 1414/1000, \dots$ which converges to $\sqrt{2}$. Then,

- $2^{\sqrt{2}}$ is the limit of $2^1, 2^{14/10}, 2^{141/100}, 2^{1414/1000}, \dots \rightarrow 2^{\sqrt{2}} = 2.665\dots$
- $3^\pi = 3^{3.14159\dots}$ is the limit of $3^3, 3^{31/10}, 3^{314/100}, 3^{3141/1000}, \dots \rightarrow 3^\pi = 31.544\dots$

Example 4.5 (negative-power of a positive number)

For a^x with $x < 0$, we use the last item in the above definition.

$$1.1^{-2} = \frac{1}{1.1^2} = \frac{1}{1.21} \quad 1.21^{-1.5} = \frac{1}{1.21^{1.5}} = \frac{1}{1.331} \quad 3^{-\pi} = \frac{1}{3^\pi} = \frac{1}{31.544\dots}$$

We have defined a^x for $a > 0$ and $x \in \mathbb{R}$. Other cases are more complicated, such as:

- a^x with $a > 0$ and $x \in \mathbb{C}$ is discussed in [♣TODO♣chapter](#) it needs a careful treatment.
- a^x with $a \leq 0$ is more complicated and requires the discussion in [♣TODO♣chapter](#). However, if x is an integer, we can safely discuss it, as follows.

Definition 4.6 (Power for negative base)

For a negative base $a < 0$, we only consider a^x for integer x .

- $a^0 := 1$.
- $a^x := a \cdot a^{x-1}$ for positive integer x .
- $a^x := 1/a^{|x|}$ for negative integer x .

We do not consider a^x if $a < 0$ and x is not an integer.

Definition 4.7 (Power for zero base)

We define $0^x := 0$ for $x > 0$, and ${}^n\sqrt{0} := 0$ for $n \in \mathbb{N}^+$. (We do not consider 0^x for $x \leq 0$.)

Be careful: Notice that we have not defined $\sqrt{x}$ for $x < 0$. Sho recommends **not** to use ${}^n\sqrt{x}$ for $x < 0$ because it is not useful for physics; it will just make you confused.

Quiz

Q4. Some of the following expressions are not defined. Find all the undefined ones.

$$\begin{array}{ccccccc} \pi^{-3} & \pi^{-1/\pi} & (-2)^0 & (-2)^{-3} & (-2)^{3/8} & -2^{-\pi} & \sqrt{-4} \\ {}^3\sqrt{3.5} & {}^{3.5}\sqrt{3} & {}^{-4}\sqrt{3} & {}^3\sqrt{0} & 0^{2.5} & 0^{-2.5} & 0.1^{0.1} \end{array}$$

In the following sections, we will only consider **powers of positive base**, i.e., a^x for $a > 0$.

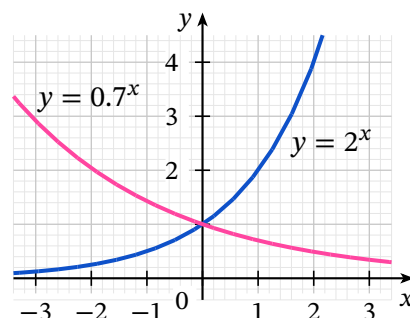
Advanced note: Technically, even if $x < 0$, we can define ${}^n\sqrt{x}$ for an odd integer n and then we can define a^x for $x \in \mathbb{Q}$ if x can be written by $x = p/q$ with an integer p and an **odd** integer q .

Advanced note: Many mathematicians define $0^0 := 1$ for some technical reasons, but we do not need it.

4.2 Exponential functions and Logarithmic functions

Powers with positive bases, such as 2^x and 0.7^x , are considered as an **exponential function** defined for $x \in \mathbb{R}$. The figure to the right shows the graphs of $y = 2^x$ and $y = 0.7^x$.

Obviously, $f(x) = a^x$ is **strictly increasing** if $a > 1$; **strictly decreasing** if $0 < a < 1$. In Section ♣TODO♣, we will check this property by calculating $f'(x)$. It is also important that a^x can take any positive real value. We will use these facts to define $\log_a x$ below in this section.



Quiz

Q5. Draw the graphs of $y = 2^x$, $y = 0.5^x$, and $y = 1^x$ by hand, without using calculators or computers.

Theorem 4.8 (Properties of Exponential Functions)

For $a > 0$, $b > 0$, and $x, y \in \mathbb{R}$,

$$a^0 = 1, \quad \frac{1}{a^x} = a^{-x}, \quad \sqrt[n]{a^x} = a^{x/n} \quad (n \in \mathbb{N}, n \geq 2); \quad (4.1)$$

$$a^x a^y = a^{x+y}, \quad \frac{a^x}{a^y} = a^{x-y}, \quad (a^x)^y = a^{xy}, \quad (ab)^x = a^x b^x, \quad \left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}. \quad (4.2)$$

Remark: a^{x^y} is usually interpreted as $a^{(x^y)}$; it is in general not equal to $(a^x)^y = a^{xy}$.

Quiz

Q6. Transform the following numbers to the form of $2^{\square}$.

- (1) 2 (2) 1024 (3) $\sqrt{2}$ (4) $\sqrt[3]{4}$ (5) 0.25 (6) $\sqrt[5]{0.25}$
 (7) 4^x (8) $(\sqrt{2})^x$ (9) $4 \cdot 2^x$ (10) $2^x/16$ (11) $2^x 4^y$ (12) $2^x/4^y$

Q7. Without a calculator, arrange the following in order from smallest to largest.

$$2, \quad 1024, \quad 2^5, \quad 8^2, \quad \sqrt{2}, \quad \sqrt[3]{4}, \quad 0.25, \quad 0.5^{0.3}, \quad \sqrt[5]{0.25}, \quad 0, \quad 1$$

Consider the equation $p = a^x$, where $a > 0$, $a \neq 1$ and $p \in \mathbb{R}$. For a given p , how many solutions x does it have? As $a \neq 1$, a^x is strictly increasing or decreasing and it takes any value $0 < a^x < \infty$. Therefore, $p = a^x$ has a unique real solution x for any $p > 0$. We call the solution $x = \log_a p$.

Definition 4.9 (Logarithm)

For $a > 0$, $a \neq 1$, and $p > 0$, we define the **logarithm** $\log_a p$ as the unique solution x of $a^x = p$.

$$\text{For } a > 0, a \neq 1, \text{ and } p > 0, \quad a^x = p \iff x = \log_a p. \quad (4.3)$$

Notice that $\log_a p$ is not defined for $p \leq 0$ or $a \leq 0$ or $a = 1$.

Quiz

Q8. Based only on the above discussion, explain why $k = \log_a(a^k)$ and $a^{\log_a k} = k$.

We here review basic properties of $\log_a x$. Drill problems are available later in this section.

Theorem 4.10 (Properties of Logarithmic Functions)

For $a > 0$ but $a \neq 1$, $b > 0$ but $b \neq 1$, $A > 0$, and $B > 0$,

$$\log_a 1 = 0, \quad \log_a a = 1, \quad \log_a A^k = k \log_a A \quad (k \in \mathbb{R}), \quad (4.4)$$

$$\log_a(AB) = \log_a A + \log_a B, \quad \log_a\left(\frac{A}{B}\right) = \log_a A - \log_a B, \quad (4.5)$$

$$\log_a A = \frac{\log_b A}{\log_b a} \quad [\text{changing the base}]. \quad (4.6)$$

4.3 Napier's number and Natural logarithm

There is a special number for the base, called **Napier's number** e :

$$e = 2.718281828\dots = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^{\infty} \frac{1}{k!} = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \frac{1}{120} + \dots \quad (4.7)$$

It is special because of the following theorem:

Theorem 4.11 (Napier's number)

$$\frac{d}{dx} e^x = e^x, \quad \frac{d}{dx} \ln x = \frac{1}{x}, \quad (4.8)$$

where the **natural logarithm** ($\ln x$) is defined by $\ln x := \log_e x$. Also, we often write e^x as $\exp(x)$.

Quiz

Q9. Draw the graphs of $y = \exp(x)$ and $y = \ln x$, using computers or calculators.

Q10. Calculate the following, where $a > 0$, $b > 0$, and $b \neq 1$.

$$(1) \frac{d}{dx} \exp(x) \quad (2) \frac{d}{dx} e^{-3x} \quad (3) \frac{d}{dx} a^x \quad (4) \frac{d}{dx} \ln(3x) \quad (5) \frac{d}{dx} \log_b x$$

Fail safe note: Review [Quiz 4.8](#) for (3). Review Eq. (4.6) for (5). Recall e is just a number.

Problems

You may skip these drill problems, as (Sho expects) you **already have learned** them.

 **4.1** Differentiate them, where a is a positive constant such that $a \neq 1$.

- | | | | | |
|----------------------|-------------------|------------------|-------------------|------------------------|
| (1) $\exp(5x)$ | (2) e^{-x^2} | (3) $3 \exp(2x)$ | (4) $e^{\sin x}$ | (5) $x(e^x)^2$ |
| (6) $\sqrt{e^x + 1}$ | (7) $e^x e^{x^2}$ | (8) $\ln(2x)$ | (9) $\ln 3x $ | (10) $\ln(x^5 + x)$ |
| (11) $\ln \cos x $ | (12) $x \ln x$ | (13) $\ln(1/x)$ | (14) $e^x \ln x$ | (15) $\ln 2x - 1 $ |
| (16) a^{4x} | (17) a^{x^2} | (18) $(2a)^{2x}$ | (19) $\log_a x^5$ | (20) $\log_a(x^5 + x)$ |

 **4.2** Simplify each expression, where a is a positive constant such that $a \neq 1$.

- | | | | |
|-------------------------|-------------------------|------------------------|-------------------------------|
| (1) $9^{-1/2}$ | (2) $16^{3/4}$ | (3) $8^{-2/3}$ | (4) $e^{3 \ln 2}$ |
| (5) $e^{\ln 3 + \ln 4}$ | (6) $e^{\ln(6+7)}$ | (7) $\ln e^{6+7}$ | (8) $e^3(e^4)^4$ |
| (9) $\log_4 8$ | (10) $\log_{10} 0.01$ | (11) $\log_{0.5} 2$ | (12) $\log_3(1/27)$ |
| (13) $\log_6 6^{10}$ | (14) $\ln 6 + \ln(1/6)$ | (15) $2 \ln 3 - \ln 9$ | (16) $\ln 12 - \ln 4 + \ln 2$ |
| (17) 1^{-a} | (18) $\log_a 1$ | (19) $\log_a(2a^2)^4$ | (20) $\log_a(\ln(e^{5a}))$ |

 **4.3** Some of the following expressions are not defined. Find all the undefined ones.

- | | | | | | | |
|-----------------|----------------|-----------------|-----------------|-----------------|-----------------|--------------------------|
| π^0 | π^{-3} | $\pi^{2.5}$ | $\pi^{-2.5}$ | π^π | $\pi^{-\pi}$ | $\pi^{-1/\pi}$ |
| 0^0 | 0^{-3} | $0^{2.5}$ | $0^{-2.5}$ | 0^π | $0^{-\pi}$ | $0^{-1/\pi}$ |
| $(-2)^0$ | $(-2)^{-3}$ | $(-2)^{2.5}$ | $(-2)^{-2.5}$ | $(-2)^\pi$ | $-2^{-\pi}$ | $-2^{-1/\pi}$ |
| $\sqrt{4.5}$ | $\sqrt{-4}$ | $\sqrt[3]{3.5}$ | $\sqrt[3.5]{3}$ | $\sqrt[4]{3}$ | $\sqrt[4]{-3}$ | $\sqrt[4]{1 + \sqrt{2}}$ |
| $\sqrt[1.5]{1}$ | $\sqrt[4]{-8}$ | $\sqrt[3]{8}$ | $\sqrt[0]{8}$ | $\sqrt[3]{\pi}$ | $\sqrt[3]{\pi}$ | $\sqrt[4]{1 - \sqrt{2}}$ |

Problems

★4.4 Solve these equations.

- | | | |
|----------------------|--------------------------|-----------------------------|
| (1) $2^{x+1} = 64$ | (2) $e^{3x-1} = 4$ | (3) $e^x = 5$ |
| (4) $\ln x - 1 = 0$ | (5) $\ln x = 4$ | (6) $\log_5(2x + 3) = 2$ |
| (7) $2^{-2x} = 7$ | (8) $\log_3(2x + 1) = 3$ | (9) $e^{2x} - 3e^x + 2 = 0$ |

*****4.5** Draw the graphs of the following functions.

- | | | | | |
|---------------|-----------------|--------------------------------------|-----------------------|-------------------------|
| (1) $y = 2^x$ | (2) $y = 0.5^x$ | (3) $y = \left(\sqrt[3]{2}\right)^x$ | (4) $y = 2 \cdot 2^x$ | (5) $y = \frac{2^x}{2}$ |
|---------------|-----------------|--------------------------------------|-----------------------|-------------------------|

Fail safe note: Draw $y = 2^x$ first. Rewrite the other functions in the form of $y = 2^\square$.

****4.6** Solve these equations.

(1) $\ln(x^2 - 3) = \ln(2x)$

(2) $\ln x + \ln(x - 1) = \ln 6$

(3) $\log_2 x + \log_2(x + 2) = 3$

(4) $\ln(x + 1) + \ln(x - 1) = \ln 3$

(5) $\log_2 x - \log_2(x - 2) = 3$

(6) $\ln(x + 1) - \ln(x - 1) = \ln 4$

(7) $\log_2|x| + \log_2|x + 2| = 3$

(8) $\ln|x + 1| + \ln|x - 1| = \ln 4$

****4.7** Calculate the derivative of x^x .

****4.8 Hyperbolic functions** are defined by

$$\cosh x := \frac{e^x + e^{-x}}{2}, \quad \sinh x := \frac{e^x - e^{-x}}{2}, \quad \tanh x := \frac{\sinh x}{\cosh x}, \quad \text{etc.} \quad (4.9)$$

(1) Calculate the derivative of $\cosh x$, $\sinh x$, and $\tanh x$.

(2) Draw the graphs of $\cosh x$, $\sinh x$, and $\tanh x$.

(3) Show the following properties:

(a) $\cosh^2 x - \sinh^2 x = 1$.

(b) $\cosh x \geq 1$ for all x .

(c) $\cosh x$ is an even function.

(d) $\sinh x$ and $\tanh x$ are odd functions.

4.4 Exponential Growth and Decay

Consider the function $f(t) = N \exp(At)$, where $N > 0$ and $A \in \mathbb{R}$.

$f(t) = N \exp(At)$ is **strictly increasing** if $A > 0$, while **strictly decreasing** if $A < 0$. (4.10)

Therefore, $f(t) = N \exp(At)$ is called **exponential growth** if $A > 0$ and **exponential decay** if $A < 0$. We learn the properties of this function in the next quiz:

Quiz

Q11. Consider the above function $f(t) = N \exp(At)$, where $N > 0$ and $A \in \mathbb{R}$.

(1) What does “strictly increasing” mean?

(2) Calculate $f'(t)$. Confirm the two statements in Eq. (4.10).

(3) Show that $f(0) = N$ and $f'(t) = Af(t)$.

Remark: We will discuss these equations in **TODO** section.

Consider an exponential decay, where the number at $t = 0$ is N . We usually express such decays by

$$\begin{aligned} f(t) &= N \exp(-\Gamma t) & (\Gamma > 0) \\ &= N \exp\left(-\frac{t}{\tau}\right) & (\text{we define } \tau := 1/\Gamma). \end{aligned} \quad (4.11)$$

Then we define the **half-life** $T_{1/2}$ as the time t such that $f(T_{1/2}) = f(0)/2$.

Quiz

- Q12.** (1) Show that $f'(t) = -\Gamma f(t)$.
(2) Show that $T_{1/2} = \tau \ln 2$.
(3) Assume $f(t_1) = 40$. Find t_2 and t_3 such that $f(t_2) = 20$ and $f(t_3) = 10$.
(4) Show that, for any t , $\frac{f(t + T_{1/2})}{f(t)} = \frac{1}{2}$.

So, each time one half-life passes, the quantity is reduced by half. If $2T_{1/2}$ has passed, the number will be 1/4 of the original number.

Problems

*****4.9** An RC circuit has charge $Q(t) = Q_0 e^{-t/\tau}$, where $\tau = RC$.

- (1) Verify that $Q(\tau) = Q_0/e$.
(2) Assume $R = 2.0 \text{ k}\Omega$ and $C = 50 \text{ }\mu\text{F}$. Calculate τ . Then, find the time t at which Q dropped to 10% of Q_0 .

Chapter 5

Vectors as arrows

Physicists understand a vector in three ways:

- an arrow in n -dimensional space ($n \in \mathbb{N}^+$, but usually $n = 3$),
- a list of n numbers arranged vertically ($n \in \mathbb{N}^+$), and
- an element of a vector space (such as a Hilbert space).

In this chapter, we will focus on the first two interpretations. The last interpretation, more abstract and mathematical, will be discussed in [♣TODO♣chap:linear-algebra](#), after we visit complex numbers.

Remark: This document discusses vectors mainly in terms of mathematics; more detailed *physical* discussion about vectors can be found in Sho's [Vector Boot Camp](#).^{#3}

Be careful: In university, we **never** use the horizontal notation $\vec{v} = (x, y, z)$ to describe vectors. Please **always** use the vertical form $\vec{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ to be prepared for [Chapter 7](#).

Remark: We use the arrow notation $\vec{v}$ because this is a first-year lecture. Usually, physicists use boldface ($\mathbf{a}, \mathbf{x}, \boldsymbol{\beta}$, etc.) to denote vectors.

Advanced note: The discussion in this chapter is only valid for *finite-dimensional* vectors because here we will define vectors as arrows in a “space”. As the “space”, readers are expected to imagine *the 3d space* of our Universe, a 2d-sheet of paper in our Universe, or something like those, and then the **dimension** of the space (defined in Theorem 5.15) will be limited to a finite integer.

5.1 What is a Vector?

You have seen vectors, such as $(5, -1)$, $(1, 0, 2)$, or $(-1, 3)$, or in the vertical form $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, etc.

However, it represents a *mathematical* nature of vectors. As a **physicist**, you have to forget about such expressions in this section (we will return to it in Section 5.7). Instead, we will first *define* vectors in a more physical way.

Definition 5.1 (Vector for physicists)

Consider a space. A **vector** is a straight arrow drawn in the space, or in general, a quantity that has both **magnitude** (= length) and **direction** in the space.

We here do not investigate what “the space” is, but you may well imagine a lecture room as the space.

^{#3}Visit <https://misho104.github.io/LecturePublic> and find gp2_boot2_vector.pdf.

Notation 5.2 (Vectors and Scalars)

We denote vectors with an arrow over the symbol, such as $\vec{v}$, $\vec{F}$, $\vec{a}$, $\vec{p}$. Their magnitudes are written as $|\vec{v}|$, $|\vec{F}|$, $|\vec{a}|$, $|\vec{p}|$, respectively. Meanwhile, a quantity that has no direction is called a **scalar**. Symbols without arrows, such as v , F , a , p , are scalars.

Be careful: v and $\vec{v}$ are **completely different objects**. They are totally unrelated, just as A and a are unrelated. However, in physics, we are sometimes *lazy* enough to write v to mean $|\vec{v}|$, the magnitude of $\vec{v}$. Please not be confused.

Quiz

Q1. Choose scalar quantities. Choose vector quantities.

- (1) $\vec{a}$ (2) p (3) $|\vec{b}|$ (4) $|g|$ (5) magnitude of $\vec{v}$
 (6) mass (7) velocity (8) speed (9) $\vec{x} + |\vec{x}|$ (10) direction of $\vec{v}$

- Q2.** (1) What do we call $|\vec{v}|$? Also, clearly write down its definition.
 (2) What do we call v ? Also, clearly write down its definition.

Fail safe note: For $x \in \mathbb{R}$, we define $|x| := \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0 \end{cases}$ and call it “the **absolute value** of x ”.

There is a special vector called the **zero vector**, which has magnitude 0 and no direction.

Definition 5.3 (Zero vector)

There is a vector with magnitude 0 and no direction. We call it **the zero vector** and denote $\vec{0}$.

Any other vectors have a direction and **positive** magnitude. Namely,

$$|\vec{0}| = 0 \quad \text{and} \quad \vec{v} \neq \vec{0} \iff |\vec{v}| > 0. \quad (5.1)$$

Quiz

Q3. Explain why $\vec{0} \neq 0$. Explain why $|\vec{0}| = 0$.

Advanced note: The uniqueness of $\vec{0}$ is easy to prove, once we clarify the definition of “=”. Namely, “ $\vec{a} = \vec{b}$ ” means “ $\vec{a}$ and $\vec{b}$ have the same magnitude and direction”. (Try to complete the proof yourself.)

5.2 Addition and Scalar multiplication

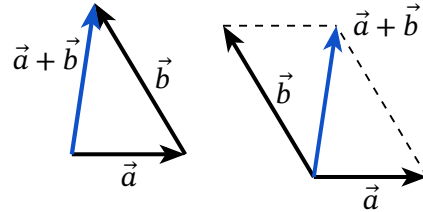
Like $+$ and $\div$ for numbers and $\wedge$ and $\vee$ for true/false, we have two operations for vectors:

- **addition:** two vectors $\vec{a}$ and $\vec{b}$ can be added; we write the result by $\vec{a} + \vec{b}$.
- **scalar multiplication:** a vector $\vec{a}$ is multiplied by a scalar k ; we write the result by $k\vec{a}$.

Definition 5.4 (Vector addition)

If $\vec{a}$ and $\vec{b}$ are vectors drawn in the same space, we can define $\vec{a} + \vec{b}$ by the vector obtained by placing the tail of $\vec{b}$ at the head of $\vec{a}$.

The triangle in the figure to the right represents this definition: the tail of $\vec{b}$ is put at the head of $\vec{a}$ to get the blue arrow $\vec{a} + \vec{b}$. You may also use the parallelogram method shown to the right: consider a parallelogram made by $\vec{a}$ and $\vec{b}$ sharing the same tail. Then its diagonal, the blue arrow, is $\vec{a} + \vec{b}$.



Notice that the two horizontal arrows in the figure are both $\vec{a}$, although their locations are different. Similarly, both blue arrows show the vector $\vec{a} + \vec{b}$. Vectors are the same if and only if they have the same magnitude and direction; location does not matter.

Definition 5.5 (Scalar multiplication)

For a vector $\vec{v}$ and a real number (scalar) k , we define $k\vec{v}$ as follows:

- if $k > 0$, $k\vec{v}$ has the magnitude $|k||\vec{v}|$ and has the same direction as $\vec{v}$.
- if $k < 0$, $k\vec{v}$ has the magnitude $|k||\vec{v}|$ and is anti-parallel to $\vec{v}$.
- if $k = 0$, $k\vec{v} = \vec{0}$.

Remark: The word “**anti-parallel**” means “in the opposite direction”. Meanwhile, **we should avoid** the ambiguous word “**parallel**” for vectors, as it may mean either the same direction or the opposite direction; use the phrase “has the same direction” instead. For more vocabulary to describe directions, please check [the Vector Boot Camp](#).

Advanced note: Watch out we assume $k \in \mathbb{R}$ in this definition.

We should carefully digest these definitions, as they are the basis for Chap♣️TODO♣️. Let’s see an example, and try the next quiz.

Example 5.6

- (1) Explain the meaning of $|k\vec{v}|$ and $|k||\vec{v}|$.
- (2) Prove $|k\vec{v}| = |k||\vec{v}|$, where $\vec{v}$ is a vector and $k \in \mathbb{R}$.
- (3) Based on the above definitions, explain what $-\vec{v}$ is.

Solution

- (1) $|k\vec{v}|$ means the magnitude of a vector $k\vec{v}$, which is a scalar multiplication of $\vec{v}$ by k . Meanwhile, $|k||\vec{v}|$ means the absolute value of k times the magnitude of $\vec{v}$.
- (2) If $k \neq 0$, then, according to the above definition, $k\vec{v}$ has the magnitude $|k||\vec{v}|$ and it means $|k\vec{v}| = |k||\vec{v}|$. If $k = 0$, then $k\vec{v} = \vec{0}$ and thus LHS = $|k\vec{v}| = 0$, while RHS = $0|\vec{v}| = 0$. ■
- (3) $-\vec{v}$ is a shorthand notation of $(-1)\vec{v}$, the scalar multiplication of $\vec{v}$ by -1 . So, $-\vec{v}$ has the same magnitude as $\vec{v}$ but is anti-parallel to $\vec{v}$.

Quiz

Q4. Find out how the following vectors are defined based on the above definitions of addition and scalar multiplication.

(1) $\vec{a} + \vec{a}$ (2) $2\vec{a}$ (3) $2\vec{a} + 3\vec{b}$ (4) $-\vec{a}$ (5) $\vec{a} - \vec{b}$ (6) $-\vec{a} - 2\vec{b}$

Addition and scalar multiplication have the following properties:

Theorem 5.7

For vectors $\vec{a}$ and $\vec{b}$ in the same space and $p, q \in \mathbb{R}$,

- | | |
|---|--|
| (A) $\vec{a} + \vec{b} = \vec{b} + \vec{a}$, | (E) $p\vec{a} + p\vec{b} = p(\vec{a} + \vec{b})$, |
| (B) $(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$, | (F) $p\vec{a} + q\vec{a} = (p + q)\vec{a}$, |
| (C) $\vec{a} + \vec{0} = \vec{a}$, | (G) $(pq)\vec{a} = p(q\vec{a})$, |
| (D) $\vec{a} + (-\vec{a}) = \vec{0}$, | (H) $1\vec{a} = \vec{a}$. |

These properties seems obvious, but in fact, they play a fundamental role in Chapter ♣TODO♣.

Problems

*****5.1** Consider a vector $\vec{a} \neq \vec{0}$ and a constant $k \in \mathbb{R}$. Consider three vectors $\vec{a}$, $k\vec{a}$, and $k^2\vec{a}$.

- (1) Which are in the same direction? Which are anti-parallel to each other?
- (2) Compare their magnitudes; which are the longest and the shortest?

*****5.2** Consider a vector $\vec{a}$ and a constant k .

- (1) Write a vector that has magnitude $|\vec{a}|$ and is anti-parallel to $\vec{a}$.
- (2) Write a vector that has magnitude $3|\vec{a}|$ and is in the same direction as $\vec{a}$.
- (3) Write a vector that has magnitude $3k|\vec{a}|$ and is in the same direction as $\vec{a}$.

Vectors with magnitude 1 are called **unit vectors**.

- (4) Write a vector that has magnitude 1 and is in the same direction as $\vec{a}$.
(Namely, write a unit vector that has the same direction as $\vec{a}$.)
- (5) Write a unit vector that is anti-parallel to $\vec{a}$.

*****5.3** Prove the next “**triangle inequality**” geometrically, i.e., only with the above definitions.

$$\text{For any two vectors } \vec{a} \text{ and } \vec{b} \text{ in the same space, } |\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|. \quad (5.2)$$

****5.4** Starting from Definition 5.4 and Definition 5.5, prove the properties in Theorem 5.7.

****5.5** Prove that, for any two vectors $\vec{a}$ and $\vec{b}$ in the same space, $||\vec{a}| - |\vec{b}|| \leq |\vec{a} - \vec{b}|$.

5.3 Inner Product

Imagine two arrows. Probably you can think the angle θ between the arrows. The angle leads you to the following **geometric** definition of the inner product.

Definition 5.8 (Inner product (geometrical definition))

For vectors $\vec{a}$ and $\vec{b}$ drawn in the same space, the **inner product** is defined by

$$\vec{a} \cdot \vec{b} := |\vec{a}| |\vec{b}| \cos \theta, \quad (5.3)$$

where θ is the angle between $\vec{a}$ and $\vec{b}$; if $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$, then $\vec{a} \cdot \vec{b} := 0$.

Quiz

Q5. Assume $\vec{a} = 3$ and $\vec{b} = 2$.

- (1) If $\vec{a} \perp \vec{b}$, then what is $\vec{a} \cdot \vec{b}$?
- (2) If $\vec{a}$ and $\vec{b}$ are anti-parallel, what is $\vec{a} \cdot \vec{b}$?
- (3) What is the maximum value of $\vec{a} \cdot \vec{b}$? When is it achieved?
- (4) If $\vec{a} \cdot \vec{b} = -3$, what can you say about the angle between $\vec{a}$ and $\vec{b}$?

Fail safe note: $\vec{a} \perp \vec{b}$ means that $\vec{a}$ and $\vec{b}$ are **perpendicular** to each other, or **normal** to each other; in other words, the angle θ between them is $\pi/2 = 90^\circ$.

This inner product has the following properties:

Theorem 5.9 (Inner product of real vectors)

For vectors $\vec{a}$ and $\vec{b}$ drawn in the same space and a constant $k \in \mathbb{R}$,

- | | |
|---|---|
| (A) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$, | (E) $\vec{a} \cdot \vec{a} \geq 0$ for any vector $\vec{a}$, |
| (B) $(\vec{a} + \vec{b}) \cdot \vec{c} = \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{c}$, | (F) $(k\vec{a}) \cdot \vec{b} = k(\vec{a} \cdot \vec{b})$, |
| (C) $\vec{a} \cdot \vec{a} = 0 \iff \vec{a} = \vec{0}$, | (G) $\vec{a} \cdot (k\vec{b}) = k(\vec{a} \cdot \vec{b})$. |
| (D) $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$, | |

Advanced note: These properties are valid only for “real vectors”. As we will see in Chapter ♣TODO♣, if $\vec{a}$ and $\vec{b}$ are “complex vectors”, then some of these properties becomes invalid and we need to modify them. In fact, all vectors in this chapter are “real vectors” because we have assumed $k \in \mathbb{R}$ in Definition 5.5.

We will skip their proof. Instead, we focus on the following geometrical properties:

Quiz

Q6. Prove the following theorems directly from Definition 5.8.

- (1) $\vec{a} \perp \vec{b} \implies \vec{a} \cdot \vec{b} = 0$.
- (2) $\vec{a} \cdot \vec{b} = 0 \implies (\vec{a} \perp \vec{b}) \vee (\vec{a} = \vec{0}) \vee (\vec{b} = \vec{0})$.
- (3) $-|\vec{a}| |\vec{b}| \leq \vec{a} \cdot \vec{b} \leq |\vec{a}| |\vec{b}|$.
- (4) $|\vec{a}|^2 = \vec{a} \cdot \vec{a}$.
- (5) $|\vec{a}| = \sqrt{\vec{a} \cdot \vec{a}}$. [Hint: Most students make mistakes in this question.]

Q7. Prove the following theorems, using Definition 5.8, Theorem 5.9, and the equations in the previous quiz.

$$(1) (\vec{a} + \vec{b}) \cdot \vec{a} = |\vec{a}|^2 + \vec{a} \cdot \vec{b}. \quad (2) |\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2.$$

Problems

*****5.6** Two vectors $\vec{a}$ and $\vec{b}$ satisfy $|\vec{a}| = 2$, $|\vec{b}| = 3$, and $\vec{a} \cdot \vec{b} = 3$. Let x, y be real numbers.

- (1) Find the angle between $\vec{a}$ and $\vec{b}$.
- (2) Calculate $|\vec{a} + \vec{b}|$.
- (3) Calculate the magnitude of $4\vec{a} + 3\vec{b}$ and $x\vec{a} + y\vec{b}$.
- (4) Explain why $|x\vec{a} + y\vec{b}|^2$ is **not** in general equal to $x^2 + y^2$.

Now let $\vec{e}_1$ and $\vec{e}_2$ satisfy $|\vec{e}_1| = |\vec{e}_2| = 1$ and $\vec{e}_1 \cdot \vec{e}_2 = 0$. Let p, q be real numbers.

- (5) Find the angle between $\vec{e}_1$ and $\vec{e}_2$.
- (6) Explain why $(p\vec{e}_1 + q\vec{e}_2) \cdot (x\vec{e}_1 + y\vec{e}_2) = px + qy$.
- (7) Explain why $|x\vec{e}_1 + y\vec{e}_2| = \sqrt{x^2 + y^2}$.

****5.7** Three vectors $\vec{A}, \vec{B}, \vec{C}$ satisfy $|\vec{A}| = 2$, $|\vec{B}| = 3$, $\vec{A} \cdot \vec{B} = 3\sqrt{2}$, and $\vec{A} \cdot \vec{C} = -1$.

- (1) Find the angle between $\vec{A}$ and $\vec{B}$.
- (2) Calculate $|\vec{A} + \vec{B}|^2$, $|\vec{A} - \vec{B}|^2$, and $|2\vec{A} + 4\vec{B}|^2$.
- (3) Calculate $(\vec{A} - 2\vec{B}) \cdot (2\vec{A} + \vec{B} + \vec{C}) + 2\vec{B} \cdot \vec{C}$.
- (4) Find k such that $|\vec{A} + k\vec{B}| = \sqrt{10}$.
- (5) Find c such that $\vec{B} + c\vec{C}$ is perpendicular to $\vec{A}$.

5.4 Cross Product (only for 3d-vectors)

For two arrows drawn in three-dimensional space, we can define the **cross product**.

Definition 5.10 (Cross product)

For $\vec{a}$ and $\vec{b}$ drawn in a *three-dimensional* space, the **cross product** $\vec{a} \times \vec{b}$ is defined as follows:

- It is a vector with magnitude $|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta$, where θ is the angle between $\vec{a}$ and $\vec{b}$.
- If $|\vec{a} \times \vec{b}| \neq 0$, we determine its direction so that $(\vec{a} \times \vec{b}) \perp \vec{a}$ and $(\vec{a} \times \vec{b}) \perp \vec{b}$.

Here, we always have two possible directions. We impose another condition to make it unique:

- If you hold the *right* hand so that your thumb points in the direction of $\vec{a}$ and your index finger in the direction of $\vec{b}$, your middle finger points in the direction of $\vec{a} \times \vec{b}$.

(This is often quoted that “the ordered triple $(\vec{a}, \vec{b}, \vec{a} \times \vec{b})$ obeys the **right-hand rule**”.)

We will not discuss much about the cross product, but only the following properties:

Theorem 5.11

For vectors $\vec{a}, \vec{b}, \vec{c}$ and real number k ,

- | | |
|---|--|
| (A) $\vec{a} \times \vec{a} = \vec{0}$. | (E) $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$. |
| (B) $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$. | (F) $\vec{a} \times \vec{b} = \vec{0}$ if $\vec{a}$ and $\vec{b}$ are parallel or anti-parallel. |
| (C) $0 \leq \vec{a} \times \vec{b} \leq \vec{a} \vec{b} $. | (G) $\vec{a} \cdot (\vec{a} \times \vec{b}) = \vec{b} \cdot (\vec{a} \times \vec{b}) = 0$. |
| (D) $(k\vec{a}) \times \vec{b} = \vec{a} \times (k\vec{b}) = k(\vec{a} \times \vec{b})$. | (H) $\vec{a} \cdot (\vec{b} \times \vec{c}) = \vec{b} \cdot (\vec{c} \times \vec{a}) = \vec{c} \cdot (\vec{a} \times \vec{b})$. |

The equation (B) is the most important. For (A), notice $\vec{a} \times \vec{a}$ is not zero.

Advanced note: Geometrical interpretation of the cross product is sometimes useful:

- $|\vec{a} \times \vec{b}|$ is the area of the parallelogram formed by $\vec{a}$ and $\vec{b}$.
- $(\vec{a} \times \vec{b}) \cdot \vec{c}$ is the volume of the parallelepiped formed by $\vec{a}, \vec{b}$, and $\vec{c}$.

Problems

★5.8 Prove (A), (B), (C), (F), and (G) of Theorem 5.11.

5.9 In physics, we **always use the **right-handed Cartesian coordinate system** to describe our three-dimensional space, which is characterized by three unit vectors $\vec{e}_x, \vec{e}_y, \vec{e}_z$ defined so that they are perpendicular to each other and $(\vec{e}_x, \vec{e}_y, \vec{e}_z)$ obeys the right-hand rule. Calculate their inner products and cross products, such as $\vec{e}_x \cdot \vec{e}_y$ and $\vec{e}_x \times \vec{e}_z$.

*5.10 Prove (B) of Theorem 5.9 and (E) of Theorem 5.11 geometrically (i.e., based on the definitions given in this chapter). Sho has his own proof but not very confident. Can you find a better proof?

5.5 Position Vectors

We have defined vectors as arrows. Arrows are not positions, so vectors are not positions. However, *once we define the origin* O in the space, we can use vectors to represent each position in the space.

Definition 5.12 (Position vector)

Consider a space and fix a point O as the origin. To each point P , we associate the vector $\vec{OP}$ and call it the **position vector** of P . We often write $\vec{p} = \vec{OP}$, $\vec{q} = \vec{OQ}$, and so on.

Consider $\vec{p} = \vec{OP}$, $\vec{p} = \vec{OQ}$, and $\vec{r} = \vec{OR}$. They are obviously dependent on the choice of the origin O . Also, the point described by $\vec{p} + \vec{q}$ will be different if we chose a different point as the origin. However, we can check that

- the vector $\vec{PQ}$ is given by $\vec{q} - \vec{p}$
- the length of the segment PQ is given by $|\vec{q} - \vec{p}|$.
- the middle point M of the segment PQ has the position vector $\vec{m} = (\vec{p} + \vec{q})/2$.

These properties are independent of the choice of O . Namely, whatever choice we did for origin, $(\vec{p} + \vec{q})/2$ represents the midpoint of PQ . For further discussion, please check [the Vector Boot Camp](#).

5.6 Intermission: Vector or Scalar or Not

Let us summarize the operations on vectors. With k a real number and $\vec{a}, \vec{b}$ vectors:

magnitude	$ \vec{a} $	$\rightarrow$ scalar
scalar multiplication	$k\vec{a}$	$\rightarrow$ vector
addition	$\vec{a} + \vec{b}$	$\rightarrow$ vector
inner product	$\vec{a} \cdot \vec{b}$	$\rightarrow$ scalar
cross product	$\vec{a} \times \vec{b}$	$\rightarrow$ vector (only in 3d)

Then, how about them? Try to ensure that you understand the meaning of each operation.

Problems

 **5.11** For each expression, answer **V** if it is a vector, **S** if scalar, and **N** if invalid (not defined). Here, $\vec{a}, \vec{b}, \dots$ are three-dimensional vectors and $a, b, \dots$ are scalars (real numbers).

- | | | | | |
|---|--|--|---|--|
| (1) $\vec{a}$ | (2) $\vec{a}^2$ | (3) $(\vec{a})^2$ | (4) $ \vec{a} ^2$ | (5) $ a ^2$ |
| (6) $\vec{0}$ | (7) $\vec{0} + 0$ | (8) $\vec{0} + \vec{0}$ | (9) $0 \cdot \vec{0}$ | (10) $\vec{0} \cdot \vec{0}$ |
| (11) $\frac{1}{\vec{a}}$ | (12) $\frac{\vec{a}}{\vec{a}}$ | (13) $\frac{\vec{a}}{b}$ | (14) $\frac{1}{ \vec{a} }$ | (15) $\frac{1}{ \vec{a} ^2}$ |
| (16) $\frac{1}{(\vec{a})^2}$ | (17) $\frac{\vec{a}}{(\vec{a} \cdot \vec{b})}$ | (18) $\frac{\vec{a}}{(\vec{a} \cdot \vec{b})^2}$ | (19) $\frac{\vec{a}}{ \vec{a} \cdot \vec{b} }$ | (20) $\frac{\vec{a}}{(\vec{a} \cdot \vec{b})}$ |
| (21) $\sqrt{\vec{a}^2}$ | (22) $\sqrt{\vec{a}}$ | (23) $\sqrt{ \vec{a} }$ | (24) $\sqrt{ a }$ | (25) $\sqrt{a^2}$ |
| (26) $a + 1$ | (27) $a + \vec{a}$ | (28) $a \times \vec{a}$ | (29) $a \cdot \vec{a}$ | (30) $a\vec{a}$ |
| (31) $\vec{a} + 1$ | (32) $\vec{a} + \vec{a}$ | (33) $\vec{a} \times \vec{a}$ | (34) $\vec{a} \cdot \vec{a}$ | (35) $\vec{a}\vec{a}$ |
| (36) a^{-2} | (37) $ \vec{a} ^{-2}$ | (38) $ \vec{a} ^{-2}$ | (39) $ \vec{a} ^{-1} \vec{a}$ | (40) $ \vec{a} ^{-1/2} \vec{a}$ |
| (41) $\vec{p} \cdot \vec{q} + \vec{p} \times \vec{q}$ | (42) $\vec{p} \times (\vec{q} \times \vec{r})$ | (43) $\vec{p} \cdot (\vec{q} \times \vec{r})$ | (44) $p(\vec{q} \times \vec{r})$ | |
| (45) $(\vec{p} \times \vec{q})^2$ | (46) $\frac{1}{ \vec{x} - \vec{r} ^{3/2}}$ | (47) $\frac{\vec{x} - \vec{r}}{ \vec{x} - \vec{r} ^{3/2}}$ | (48) $\frac{\vec{d} \times (\vec{x} - \vec{r})}{ \vec{x} - \vec{r} ^{3/2}}$ | |
| (49) $\frac{d\vec{y}}{dx}$ | (50) $\frac{dy}{d\vec{x}}$ | | | |

5.7 Axes and Components

So far, we considered vectors as arrows drawn in some space. Vectors are characterized only by the magnitude and direction. Now, we are going to *recall* the component-wise notation such as $\vec{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$. We define “orthonormal basis vectors” and then see we *can* describe an arrow by a list of numbers.

Definition 5.13 (Orthonormal basis vectors)

If n vectors $\vec{e}_1, \dots, \vec{e}_n$ satisfy the following properties, we call them **orthonormal basis**:

- All of them are unit vectors, i.e., $|\vec{e}_i| = 1$ for all i .
- Any of them are perpendicular, i.e., $i \neq j \implies \vec{e}_i \cdot \vec{e}_j = 0$.
- We cannot add any more vectors without violating the above two rules.

Precisely speaking, we call the set $\{\vec{e}_1, \dots, \vec{e}_n\}$ “an orthonormal basis”. Then, if we fix *one* orthonormal basis to use, we call its members “orthonormal basis vectors”.

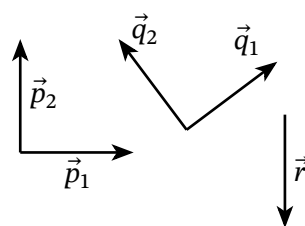
Fail safe note: A woman, two women. A nucleus, two nuclei. A basis, two bases. A matrix, two matrices.

Example 5.14 (Basis vectors for arrows on this sheet)

Let’s limit our “space” to this page and consider the five vectors drawn to the right. Assume their length are all “1”.

If we choose $\vec{p}_1$ and $\vec{p}_2$, they form *one* orthonormal basis $\{\vec{p}_1, \vec{p}_2\}$ because they are unit vectors, perpendicular to each other, and we cannot add any more. Similarly, $\{\vec{q}_1, \vec{q}_2\}$ is *another* orthonormal basis.

Meanwhile, $\{\vec{p}_1, \vec{q}_2\}$ and $\{\vec{p}_1, \vec{p}_2, \vec{q}_1, \vec{q}_2\}$ are not orthonormal bases because the members are not orthogonal. $\{\vec{p}_1\}$ is not an orthonormal basis because we can add one more vector, such as $\vec{r}$, without breaking the conditions.



Quiz

Q8. Consider the figure in the above example. Which sets are orthogonal bases?

$$\{\vec{p}_1, -\vec{p}_2\}, \quad \{\vec{p}_1, \vec{r}\}, \quad \{\vec{p}_2, \vec{r}\}, \quad \{\vec{p}_1, \vec{p}_2, \vec{r}\}, \quad \{\vec{q}_1, 2\vec{q}_2\}, \quad \{-\vec{q}_1, -\vec{q}_2\}.$$

A space has infinitely many orthonormal bases; we can choose an orthonormal basis at our convenience. However, the number of the members is fixed by the space we considered.

Theorem 5.15 (Properties of orthonormal basis vectors)

- Any orthonormal bases of a space have the same number of vectors. We call the number **the dimension** of the space.
- If $\{\vec{e}_1, \vec{e}_2, \dots, \vec{e}_n\}$ is an orthonormal basis, any arrow $\vec{v}$ in the space can be expressed as

$$\vec{v} = c_1 \vec{e}_1 + c_2 \vec{e}_2 + \dots + c_n \vec{e}_n = \sum_{k=1}^n c_k \vec{e}_k, \quad \text{where } c_k \in \mathbb{R} \quad (5.4)$$

and this expression is *unique*, i.e., if $\vec{v}$ is expressed by

$$\begin{aligned} \vec{v} &= c_1 \vec{e}_1 + c_2 \vec{e}_2 + \dots + c_n \vec{e}_n \\ &= d_1 \vec{e}_1 + d_2 \vec{e}_2 + \dots + d_n \vec{e}_n, \end{aligned}$$

then all the coefficients are equal: $c_k = d_k$.

Quiz

Q9. Show that the numbers c_k in Eq. (5.4) are actually determined by $c_k = \vec{v} \cdot \vec{e}_k$.

Advanced note: Sho thinks this theorem is not difficult to prove because we only think finite-dimensional spaces, but there can be caveats or subtleties. A more rigorous construction is in Chapter ♣TODO♣.

If you choose an orthogonal basis, then it automatically defines the **axes** of the space:

- For a 2d space, we call the directions of the basis vectors as *x*-axis and *y*-axis, respectively. Sho usually writes the basis vectors by $\vec{e}_x$ and $\vec{e}_y$, but other textbooks may write as $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$.
- For a 3d space, we call the directions of the basis vectors as *x*-axis, *y*-axis, and *z*-axis, respectively. The basis vectors are expressed by $(\vec{e}_x, \vec{e}_y, \vec{e}_z)$ or $(\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}})$. Here, physicists **always** choose the axes so that $(\vec{e}_x, \vec{e}_y, \vec{e}_z)$ obeys the **right-hand rule** (see Section 5.4).

They are called **2d Cartesian coordinate system** and **3d right-handed Cartesian coordinate system**, respectively. In general, a coordinate system defined by an orthonormal basis is called Cartesian coordinate system.

Now we are ready to express vectors in their **components** because we have reached Eq. (5.4); if we *fix* an orthonormal basis, any vectors are expressed as Eq. (5.4) with *uniquely determined* numbers $c_k := \vec{e}_k \cdot \vec{v}$.

Definition 5.16 (Components of a vector)

If we fix an orthonormal basis and label the basis vectors by $\vec{e}_1, \dots, \vec{e}_k$, any vector $\vec{v}$ can be written as

$$\vec{v} = c_1 \vec{e}_1 + \dots + c_k \vec{e}_k, \quad c_k := \vec{e}_k \cdot \vec{v} \in \mathbb{R}.$$

We call c_k **the *k*-th component** of $\vec{v}$ and express $\vec{v}$ by them: $\vec{v} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_k \end{pmatrix} = \begin{pmatrix} \vec{e}_1 \cdot \vec{v} \\ \vec{e}_2 \cdot \vec{v} \\ \vdots \\ \vec{e}_k \cdot \vec{v} \end{pmatrix}.$

Advanced note: We have to make sure this representation is *well-defined*; we do not want to have two different expressions for one vector, or two different vectors having the same expressions.

Because c_k is uniquely determined, the component-wise notation is unique for a vector. Meanwhile, if $\vec{a}$ and $\vec{b}$ are different but have the same component-wise notation, it means $\vec{a} - \vec{b} =: \vec{\Delta}$ has the same notation as $\vec{0}$ (why?). It means $\vec{\Delta}/|\vec{\Delta}|$ is a unit vector orthogonal to all of $\vec{e}_k$, which contradicts that $\{\vec{e}_1, \dots, \vec{e}_n\}$ is the basis (why?). Accordingly, different vectors must have different component-wise notation.

Vectors in n -dimensional spaces are called **n -dimensional vectors**. Since an orthonormal basis in a n -dimensional space has n basis vectors, a n -dimensional vectors are expressed with n real numbers $c_1, \dots, c_n$.

Example 5.17

Prove the following equations in a 2d space.

$$(1) \begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} a+p \\ b+q \end{pmatrix} \quad (2) \begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} p \\ q \end{pmatrix} = ap + bq \quad (3) \left| \begin{pmatrix} a \\ b \end{pmatrix} \right| = \sqrt{a^2 + b^2}$$

Solution

(1) Since $\begin{pmatrix} a \\ b \end{pmatrix}$ means $a\vec{e}_x + b\vec{e}_y$ and $\begin{pmatrix} p \\ q \end{pmatrix}$ means $p\vec{e}_x + q\vec{e}_y$ under some orthonormal basis $(\vec{e}_x, \vec{e}_y)$,

$$\text{LHS} = (a\vec{e}_x + b\vec{e}_y) + (p\vec{e}_x + q\vec{e}_y) = (a+p)\vec{e}_x + (b+q)\vec{e}_y = \text{RHS},$$

where we used the equations in Theorem 5.7. ■

(2) Similarly, using the equations in Theorem 5.9 and Definition 5.13,

$$\begin{aligned} \text{LHS} &= (a\vec{e}_x + b\vec{e}_y) \cdot (p\vec{e}_x + q\vec{e}_y) \\ &= ap(\vec{e}_x \cdot \vec{e}_x) + bp(\vec{e}_y \cdot \vec{e}_x) + aq(\vec{e}_x \cdot \vec{e}_y) + bq(\vec{e}_y \cdot \vec{e}_y) = ap + bq. \blacksquare \end{aligned}$$

(3) Because of (2), $(\text{LHS})^2 = \begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} a \\ b \end{pmatrix} = a^2 + b^2$. Since $(\text{LHS}) \geq 0$, $\text{LHS} = \sqrt{a^2 + b^2} = \text{RHS}$. ■

Fail safe note: In (3), the phrase “ $\text{LHS} \geq 0$ ” is necessary. Without it, you can only claim “ $= \pm\sqrt{a^2 + b^2}$ ”.

Notice that we *proved* these equations based on Theorem 5.7 etc. These equations are not *definitions* or *assumptions*, but *derived statements* ($\rightarrow$ Section 3.5). So, we also have to prove a few more statements.

Quiz

Q10. Consider a 2d space.

(1) Prove $\vec{0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\vec{e}_x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, and $\vec{e}_y = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. [Hint: Recall $c_k = \vec{e}_k \cdot \vec{v}$.]

(2) Prove $k \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ka \\ kb \end{pmatrix}$, where $k \in \mathbb{R}$.

Problems

*****5.12** Consider a 3d space. An orthonormal basis $(\vec{e}_x, \vec{e}_y, \vec{e}_z)$ is taken according to the right-

hand rule. Let $\vec{a} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ and $\vec{p} = \begin{pmatrix} p \\ q \\ r \end{pmatrix}$.

(1) Prove $\vec{0} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$, $\vec{e}_x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\vec{e}_y = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, and $\vec{e}_z = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

(2) Prove $k\vec{a} + l\vec{b} = \begin{pmatrix} ka + lp \\ kb + lq \\ kc + lr \end{pmatrix}$.

(3) Prove $\vec{a} \cdot \vec{p} = ap + bq + cr$ and $|\vec{a}| = \sqrt{a^2 + b^2 + c^2}$.

(4) Express $\vec{a} \times \vec{p}$ with using a, b, c, p, q, r .

****5.13** Consider a n -dimensional space ($n \in \mathbb{N}^+$) and an orthonormal basis $(\vec{e}_1, \dots, \vec{e}_n)$ of it. Prove the following.

- $\vec{0} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$ and $(i\text{-th component of } \vec{e}_j) = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$
- Consider $p, q \in \mathbb{R}$ and n -dimensional vectors $\vec{a}$ and $\vec{b}$. Let the k -th component of $\vec{a}$ be a_k and the k -th component of $\vec{b}$ be b_k . Then, the k -th component of $p\vec{a} + q\vec{b}$ is equal to $pa_k + qb_k$. Also, $\vec{a} \cdot \vec{b} = \sum_{k=1}^n a_k b_k$ and $|\vec{a}| = \sqrt{\sum_{k=1}^n a_k^2}$.

Be careful: Again, these formulas are *consequences* of the definitions we gave earlier. They are **not** new definitions.

Since you must be familiar with this component-wise notation and calculations based on it, we do not discuss it further. Consult your high-school math textbook or Sho's [Vector Boot Camp](#) if you are not confident in such calculations.

Vectors are closely related to matrices, the main topic of **♣TODO♣**. We will there discuss both real and complex matrices, and thus complex vectors will be introduced there. Since complex vectors are not arrows, we need to *define* complex vectors in a way different from what we did in this chapter.

To prepare for the discussion on complex matrices in **♣TODO♣**, we first review **complex numbers** in the next chapter.

Problems

***5.14** Complete the proof in “Advanced Note” after Definition 5.16.

***5.15** When we define a Cartesian coordinate system for a 3d space, we require that $(\vec{e}_x, \vec{e}_y, \vec{e}_z)$ satisfies the right-hand rule. However, similar rules are not required for 2d spaces. Why?

Chapter 6

Complex Numbers

You have learned complex numbers in high school but it is worth reviewing them in a university style. As you learned in Section 3.5, try to distinguish between definitions and derived consequences.

Remark: Geometrical discussions, such as **complex planes**, are not included in this document, mainly because you are expected to be familiar with it.

6.1 Complex Numbers

Definition 6.1 (Complex numbers)

A **complex number** is given by the form

$$z = a + bi, \text{ where } a, b \in \mathbb{R} \text{ and } i \text{ is a symbol satisfying } i^2 = -1. \quad (6.1)$$

We call the newly-introduced symbol i **imaginary unit**. Also, a is called the **real part** of z and b the **imaginary part** of z , written by

$$\operatorname{Re}(z) := a, \quad \operatorname{Im}(z) := b.$$

The set of all complex numbers is written by $\mathbb{C}$.

Be careful: $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$ are both real. If $z = 3 + 5i$, then $\operatorname{Im}(z) = 5$ (not $5i$).

Remark: We use i to denote the imaginary unit. You may see other notations such as i , j , j , or $\sqrt{-1}$. However, Sho recommends you **not** to write $\sqrt{-1}$ since it may make you confused. Recall, in [Chapter 4](#), we *avoided* definitions of $\sqrt{x}$ for $x < 0$.

Advanced note: Notice that i is just a symbol and it has only one property $i \cdot i = -1$. You may well ask “ $-i$ also satisfies $(-i)^2 = -1$; which should we choose as i ?” but it sounds a weird question since “ $-i$ ” was not *a priori*: it was not there before we introduced i , so we can discuss $-i$ only after we fix i .

Hereafter, when we write $z = a + bi \in \mathbb{C}$, we implicitly assume a and b are real numbers.

Definition 6.2

Consider $z = a + bi \in \mathbb{C}$.

- z is a real number if $\operatorname{Im} z = b = 0$. Namely, if $b = 0$, then $z = a + 0i = a$.
- z is called **pure-imaginary** if $\operatorname{Re} z = a = 0$ but $z \neq 0$, namely, if $z = bi$ with $b \neq 0$.

Also, for another complex number $c + di$, we define $a + bi = c + di \iff a = c \wedge b = d$.

It is debatable if 0 is pure-imaginary or not, so you need to be careful when you discuss 0 (but usually 0 is not included in pure-imaginary numbers.) Some people write the set of all pure-imaginary number by $i\mathbb{R}$, but then you need to pay attention to this issue.

For complex numbers $z_1 = a_1 + b_1i$ and $z_2 = a_2 + b_2i$, addition and multiplication are defined by

$$\begin{aligned} z_1 + z_2 &= (a_1 + a_2) + (b_1 + b_2)i, & z_1 z_2 &= (a_1 + b_1i)(a_2 + b_2i) \\ & & &= (a_1 a_2 + b_1 b_2 i^2) + (a_1 b_2 + a_2 b_1)i \\ & & &= (a_1 a_2 - b_1 b_2) + (a_1 b_2 + a_2 b_1)i. \end{aligned}$$

Furthermore, two more important operations are defined:

Definition 6.3 (Complex conjugate and Absolute value)

For $z = a + bi \in \mathbb{C}$, we define

- its **complex conjugate** by $\bar{z} := a - bi$,
- its **absolute value** by $|z| := (\bar{z} z)^{1/2} = (a^2 + b^2)^{1/2}$.

Quiz

- Q1.** Prove that, if $z = a + bi$, then $\bar{z}z$ is a real and positive number.
- Q2.** Explain why $|z| \geq 0$. [Hint: See the definition carefully. What does $x^{1/2}$ mean?]
- Q3.** For each z below, find $\text{Re}(z)$, $\text{Im}(z)$, $\bar{z}$, and $|z|$.
- $$z = 2 + 7i, \quad z = -3 - i, \quad z = 5, \quad z = -4i, \quad z = 0.$$
- Q4.** Find $x, y \in \mathbb{R}$ such that $(x + 1) + (2y - 3)i = 4 - i$.

Problems

 **6.1** [Note: This is a timed practice; just write down the answer, without explanation.] Consider the following numbers. Assume $a, b, k \in \mathbb{R}$.

$$\begin{aligned} &-2 + i, \quad 1 - i, \quad 3 + 4i, \quad -2(3 + 4i), \quad -2, \quad -2i, \quad -9i, \quad 0, \quad i, \quad 4, \\ &\sqrt{2} + i, \quad -\sqrt{3} + 2i, \quad 3 + i\sqrt{5}, \quad -1 - i\sqrt{7}, \quad -\sqrt{11} + i\sqrt{14}, \\ &\frac{i + \sqrt{5}}{\sqrt{2}}, \quad \frac{i - \sqrt{5}}{\sqrt{2}}, \quad \frac{\sqrt{3}}{\sqrt{2}} + \frac{\sqrt{3}}{\sqrt{2}}i, \quad -\frac{\sqrt{2}}{2} + \frac{\sqrt{6}}{2}i, \quad a + 2i, \quad a - bi, \quad ki. \end{aligned}$$

- (1) What are the real part, imaginary part, and complex conjugate of each?
- (2) What are their absolute values? (goal time: 120 seconds)

 **6.2** Calculate the following.

- | | | |
|----------------------------|-----------------------------------|------------------------------------|
| (1) $(3 + 2i) + (4 - 5i)$ | (2) $(-2 + 7i) - (5 + 3i)$ | (3) $(6 - 4i) - (-1 + 2i)$ |
| (4) $(2 + 3i)(4 - i)$ | (5) $(-1 + 5i)(2 + 3i)$ | (6) $(3 - 2i)(3 + 2i)$ |
| (7) $i(3 + i) - i(1 - 4i)$ | (8) $(1 - 2i) + \overline{4 + i}$ | (9) $\overline{(3 - 4i) + 1 + 2i}$ |
| (10) $(1 + 2i)^4$ | (11) $(1 - 2i)^4$ | (12) $(2 + 3i)^3(2 - 3i)^3$ |
| (13) $\frac{2 + i}{-i}$ | (14) $\frac{11 + 4i}{1 + 2i}$ | (15) $\frac{1 - i}{(1 + i)^2}$ |

Fail safe note: A fraction w/z of complex numbers can be simplified by multiplying both numerator and denominator by $\bar{z}$:

$$\frac{w}{z} = \frac{w\bar{z}}{z\bar{z}} = \frac{w\bar{z}}{|z|^2}. \quad \text{For example, } \frac{11 + 4i}{1 + 2i} = \frac{(11 + 4i) \times (1 - 2i)}{(1 + 2i) \times (1 - 2i)} = \frac{19 - 18i}{5}.$$

****6.3** Find $z \in \mathbb{C}$ that satisfies $z^2 = 1 + 2i$.

As we already saw some examples, powers of complex numbers are defined, but **only for** $n \in \mathbb{Z}$.

- $\alpha^1 = \alpha, \quad \alpha^2 = \alpha\alpha, \quad \alpha^3 = \alpha^2\alpha = \alpha\alpha\alpha, \dots$
- For $\alpha \neq 0, \quad \alpha^0 = 1, \quad \alpha^{-1} = \frac{1}{\alpha}, \quad \alpha^{-2} = \frac{1}{\alpha^2}, \quad \dots$

In particular, $(a + bi)^{-n} = \frac{1}{(a + bi)^n} = \frac{1}{(a + bi)^n} \times \frac{(a - bi)^n}{(a - bi)^n} = \frac{(a - bi)^n}{(a^2 + b^2)^n} \quad \text{if } a + bi \neq 0.$



The following relations hold for these operations. They are somewhat obvious, but with using them you can speed up your calculations.

Theorem 6.4 (Properties of Complex-number arithmetic)

For $\alpha, \beta, \gamma \in \mathbb{C}$ and $n, m \in \mathbb{Z}$,

$$\begin{aligned} (\alpha + \beta) + \gamma &= \alpha + (\beta + \gamma), & \alpha + \beta &= \beta + \alpha, & 0 + \alpha &= \alpha, \\ (\alpha\beta)\gamma &= \alpha(\beta\gamma), & \alpha\beta &= \beta\alpha, & 1\alpha &= \alpha, & \alpha(\beta + \gamma) &= \alpha\beta + \alpha\gamma. \end{aligned} \quad (6.2)$$

$$\alpha^n \alpha^m = \alpha^{n+m}, \quad (\alpha^n)^m = \alpha^{nm}, \quad (\alpha\beta)^n = \alpha^n \beta^n, \quad \alpha^{-1} = \frac{\bar{\alpha}}{|\alpha|^2}, \quad \left(\frac{\alpha}{\beta}\right)^n = \frac{\alpha^n}{\beta^n}; \quad (6.3)$$

$$\overline{\bar{\alpha}} = \alpha, \quad \overline{\alpha + \beta} = \bar{\alpha} + \bar{\beta}, \quad \overline{\alpha\beta} = \bar{\alpha} \bar{\beta}, \quad \overline{\alpha^n} = (\bar{\alpha})^n, \quad \overline{\left(\frac{\alpha}{\beta}\right)} = \frac{\bar{\alpha}}{\bar{\beta}}; \quad (6.4)$$

$$|\bar{\alpha}| = |\alpha|, \quad |\alpha\beta| = |\alpha||\beta|, \quad \left|\frac{\alpha}{\beta}\right| = \frac{|\alpha|}{|\beta|}, \quad |\alpha^n| = |\alpha|^n, \quad (6.5)$$

$$|\alpha| - |\beta| \leq |\alpha - \beta|, \quad |\alpha + \beta| \leq |\alpha| + |\beta|, \quad (6.6)$$

$$\operatorname{Re}(z) = \frac{z + \bar{z}}{2}, \quad \operatorname{Im}(z) = \frac{z - \bar{z}}{2i}. \quad (6.7)$$

Example 6.5

Express following expressions without “Re” and “Im”, i.e., only with z and $\bar{z}$.

1. $\operatorname{Re}(z)^2$ 2. $\operatorname{Re}(z^2)$ 3. $\operatorname{Re}(z) \operatorname{Im}(z)$

Solution

$$\operatorname{Re}(z)^2 = \left(\frac{z + \bar{z}}{2}\right)^2 = \frac{(z + \bar{z})^2}{4}; \quad \operatorname{Re}(z^2) = \frac{z^2 + \bar{z}^2}{2} = \frac{z^2 + \bar{z}^2}{2}; \quad \operatorname{Re}(z) \operatorname{Im}(z) = \frac{z^2 - \bar{z}^2}{4i}.$$

Quiz

Q5. Calculate them.

$$(1) \left|(\sqrt{2} + i\sqrt{7})^4\right| \quad (2) |11 + 22i|^2 \quad (3) \left|\frac{1+i}{1-i}\right| \quad (4) (1+i)\overline{(1+i)^2}$$

Q6. Express following expressions without “Re” and “Im”, i.e., only with z and $\bar{z}$.

$$(1) (\operatorname{Im}(z))^2 \quad (2) \operatorname{Im}(2z) \quad (3) \operatorname{Re}(z) + \operatorname{Im}(z)$$

Problems

★6.4 Let $z = 2 + i$ and $w = 1 - 2i$. Calculate:

$$\begin{array}{lllll} (1) z + w & (2) \bar{z} & (3) \bar{w} & (4) |z| & (5) |w| \\ (6) \bar{z} + \bar{w} & (7) \overline{z + w} & (8) z\bar{w} & (9) \bar{z}w & (10) |zw| \\ (11) \left|\frac{z}{w}\right| & (12) |z + w| & (13) |z| + |\bar{w}| & (14) |zw + z\bar{w}| & (15) |\bar{z}w + z\bar{w}| \end{array}$$

★6.5 Calculate i^2 , i^4 , and i^{127} . Also, for $z = (1 + i)/\sqrt{2}$, calculate z^{127} .

★6.6 Prove the following statements, where $z, \alpha, \beta \in \mathbb{C}$.

(1) $\overline{z_1 z_2} = \overline{z_1} \overline{z_2}$.

(2) $|z_1 z_2| = |z_1| |z_2|$.

(3) $\text{Im}(z) = (z - \bar{z})/(2i)$.

**6.7 Prove that, for $a, b, c \in \mathbb{R}$ and $z \in \mathbb{C}$, the solution of $az^2 + bz + c = 0$ is given by

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad \text{if } b^2 \geq 4ac; \quad z = \frac{-b \pm i\sqrt{|b^2 - 4ac|}}{2a} \quad \text{if } b^2 < 4ac. \quad (6.8)$$

**6.8 Find all complex numbers z satisfying the following equations.

(1) $z^2 + 16 = 0$

(2) $z^2 + 4z + 5 = 0$

(3) $z^2 + 16i = 0$

(4) $z^2 + 4z + 5i = 0$

**6.9 Let's prove $0 \leq |\alpha + \beta| \leq |\alpha| + |\beta|$ step by step, where $\alpha, \beta \in \mathbb{C}$.

(1) For $z \in \mathbb{C}$, prove the next: $z \neq 0 \implies z\bar{z} > 0$. [Hint: Let $z = a + bi$.]

(2) For $z \in \mathbb{C}$, prove $|z| \geq \text{Re}(z)$. Then, let $z = \alpha\bar{\beta}$ and simplify $2|z| - 2\text{Re}(z)$.

(3) Calculate $(|\alpha| + |\beta|)^2 - |\alpha + \beta|^2$ and prove it is not negative (≥ 0).

(4) Prove $0 \leq |\alpha + \beta| \leq |\alpha| + |\beta|$. Also, find the condition for $|\alpha + \beta| = |\alpha| + |\beta|$.

**6.10 Let $p(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0$ be a polynomial with *real* coefficients ($a_k \in \mathbb{R}$). Prove that if $z_0 \in \mathbb{C}$ is a solution of $p(z) = 0$, then so is $\bar{z}_0$.

6.2 Polar Form

Let us begin with a theorem.

Theorem 6.6 (Polar form)

A complex number $z = a + bi$, where $a, b \in \mathbb{R}$, can be expressed in the following **polar form**:

$$z = r(\cos \theta + i \sin \theta), \quad \text{where } r = |z| \text{ is the absolute value of } z, \quad (6.9)$$

$\theta \in \mathbb{R}$ is called the **argument** of z .

For $z \neq 0$, this expression is unique up to the 2π -periodicity of θ .

Proof: Let $z = a + bi$ with $a, b \in \mathbb{R}$. For $z = 0$, the statement is valid because $r = |z| = 0$. Now we assume $z \neq 0$ and, with $r := |z|$, consider a point P at $(a/r, b/r)$ on xy -plane. Since $(a/r)^2 + (b/r)^2 = 1$, P is on the unit circle around the origin O and thus can be expressed by $(\cos \theta, \sin \theta)$ with θ being the angle from the positive x -axis to $\overrightarrow{OP}$ and the choice of θ is unique up to the 2π -periodicity. It means $a/r = \cos \theta$ and $b/r = \sin \theta$, i.e., $z = r \cos \theta + ir \sin \theta$. ■

Advanced note: In this document, the **trigonometric functions** $\cos \theta$ and $\sin \theta$ are defined using the unit circle in xy -plane and this proof employs that definition. In advanced mathematics, we often take another approach, where we first define e^x for $x \in \mathbb{R}$ by a **power series**. Then we extend e^x to $\mathbb{C}$ and, from this complex function e^z , we define $\cos x$ and $\sin x$ for $x \in \mathbb{R}$ by Euler's formula. See ♠TODO♠Lorem ipsum dolor sit. for further discussion.

Quiz

Q7. Find the complex number expressed by the following polar form:

- (1) $r = \sqrt{2}, \theta = \pi/4$ (2) $r = 2, \theta = -\pi/6$ (3) $r = 4, \theta = \pi/2$
 (4) $r = \sqrt{2}, \theta = 9\pi/4$ (5) $r = 2, \theta = -5\pi/6$ (6) $r = 4, \theta = -\pi$

Similar to $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$, we may consider a function $\operatorname{Arg}(z)$ to return its argument. Here, however, we need to care the **multivaluedness** in θ . For example, (1) and (4) in the above quiz express the same number $z = 1 + i$, so there can be multiple values of θ for $z = 1 + i$ (and, in fact, for any other numbers). A convenient choice is to define $-\pi < \operatorname{Arg}(z) \leq \pi$:

Definition 6.7

For $z \in \mathbb{C}, z \neq 0$,

$$\operatorname{Arg}(z) := \text{the argument of } z, \text{ chosen between } -\pi \text{ (exclusive) and } \pi \text{ (inclusive)}. \quad (6.10)$$

We often call it “we choose the **principle value** of the argument of z so that $-\pi < \operatorname{Arg}(z) \leq \pi$ ”.

Quiz

Q8. Find $\operatorname{Arg}(5i)$, $\operatorname{Arg}(2 + 2i)$, $\operatorname{Arg}(-1 - i)$, and $\operatorname{Arg}(1 - i)$.

6.3 Euler's formula

So far, we have seen $z + w$, zw , and z^n for $n \in \mathbb{Z}$, very similar to the real numbers, and $\bar{z}$ and $|z|$. Next, we want to define e^z for $z \in \mathbb{C}$. Here, we want to keep the key properties of e^x such as $de^{ax}/dx = ae^{ax}$ and $e^{a+b} = e^a e^b$ even for $a, b \in \mathbb{C}$. So,

Definition 6.8

We **define**, for a complex number $z = a + bi$ ($a, b \in \mathbb{R}$),

$$e^{a+bi} := e^a (\cos b + i \sin b). \quad (6.11)$$

$$\text{In particular, for } \theta \in \mathbb{R}, \quad e^{i\theta} = \cos \theta + i \sin \theta \quad (\text{Euler's formula}). \quad (6.12)$$

Advanced note: Let $a, b, x \in \mathbb{R}$. Then, we want to keep $e^{(a+bi)x} = e^{ax} e^{bix}$ and thus we just have to define $e^{i\theta}$ for a real number $\theta = bx$. Letting $e^{i\theta} = f(\theta) + ig(\theta)$, the requirement of derivative is given by $de^{ibx}/dx = ibe^{ibx}$, where the LHS is $b \cdot de^{i\theta}/d\theta = bf'(\theta) + ibg'(\theta)$ and the RHS is $ib(f(\theta) + ig(\theta))$. It means $f'(\theta) = -g(\theta)$ and $g'(\theta) = f(\theta)$. Together with $f(0) = 1$ and $g(0) = 0$, we get $f(\theta) = \cos(\theta)$ and $g(\theta) = \sin(\theta)$ and now we have understand our wanted definition is $e^{a+ib} = e^a (\cos b + i \sin b)$.

Accordingly, the polar form of complex numbers is given by

$$z = a + bi = r(\cos \theta + i \sin \theta) = r \exp(i\theta) = |z| \exp(i \operatorname{Arg}(z)). \quad (6.13)$$

As we defined e^z so that the key properties are kept, the following formulae are valid. (Compare with Theorem 4.8!).

Theorem 6.9 (Properties of Exponential Functions (2))

$$\text{For } z, w \in \mathbb{C}, \quad \frac{1}{e^z} = e^{-z}, \quad e^z e^w = e^{z+w}, \quad \frac{e^z}{e^w} = e^{z-w}, \quad (e^z)^w = e^{zw}. \quad (6.14)$$

Because $\ln a$ is already defined for $a > 0$, we can use it to consider a^z (but now only for $a > 0$):

$$\text{For } z \in \mathbb{C} \text{ and } a > 0, \quad a^z = (e^{\ln a})^z = e^{z \ln a}. \quad (6.15)$$

We will ♠TODO♠later? discuss a^z for non-positive a , such as $a = -1$ or i .

Quiz

Q9. Express the following number in the form $a + bi$:

- | | | | | | |
|-------------------|--------------|---------------|-----------------|----------------|----------------|
| (1) e^i | (2) e^{2i} | (3) e^{-i} | (4) e^{1+i} | (5) $(e^i)^4$ | (6) e^{i^4} |
| (7) $e^{i \ln 3}$ | (8) 2^i | (9) 2^{2+i} | (10) 2^{2+2i} | (11) 4^{1+i} | (12) 5^{i^2} |

The next theorem is extremely important:

Theorem 6.10

$$\text{For } \theta \in \mathbb{R}, \quad \cos \theta = \operatorname{Re}(e^{i\theta}) = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \operatorname{Im}(e^{i\theta}) = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \quad (6.16)$$

Then for $\theta \in \mathbb{R}$ and $n \in \mathbb{Z}$, de Moivre's theorem $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$ holds, but we do not need to memorize it. Just use the equation $e^{in\theta} = (e^{i\theta})^n$, e.g.,

Example 6.11

Express $\cos 4\theta$ in terms of $\cos \theta$.

Solution

Write $c = \cos \theta$ and $s = \sin \theta$ for brevity. Then.

$$\cos 4\theta = \operatorname{Re}(e^{4i\theta}) = \operatorname{Re}(e^{i\theta})^4 = \operatorname{Re}(c + is)^4 = \operatorname{Re}(c^4 + 4c^3is + 6c^2i^2s^2 + 4ci^3s^3 + i^4s^4).$$

As $s^2 = 1 - c^2$, we obtain $\cos 4\theta = c^4 - 6c^2(1 - c^2) + (1 - c^2)^2 = 8\cos^4 \theta - 8\cos^2 \theta + 1$.

Quiz

Q10. Using the above properties, express $\cos 3\theta$, $\sin 3\theta$, $\cos(\theta + \varphi)$, and $\sin(\theta + \varphi)$ in terms of $\cos \theta$, $\sin \theta$, $\cos \varphi$, and $\sin \varphi$.

Fail safe note: Expand both sides of the equality $e^{i(\theta+\varphi)} = e^{i\theta}e^{i\varphi}$.

Problems

 **6.11** Compute the following and express in the form $a + bi$.

- | | | | |
|---------------------------|--------------------------------|-----------------|------------------------|
| (1) $e^{i\pi/6}$ | (2) $e^{i\pi}$ | (3) $e^{2i\pi}$ | (4) $2e^{-i\pi/3}$ |
| (5) $\sqrt{2}e^{-i\pi/4}$ | (6) $e^{i\pi/2} + e^{-i\pi/2}$ | (7) $(1 + i)^4$ | (8) $(\sqrt{3} + i)^6$ |

★6.12 Convert to polar form $re^{i\theta}$, choosing $\operatorname{Arg}(z) \in (-\pi, \pi]$.

- | | | | |
|---------------------|---------------------|----------|---------------------|
| (1) $1 + \sqrt{3}i$ | (2) -2 | (3) $-i$ | (4) $3 - 3i$ |
| (5) $-1 + i$ | (6) $-\sqrt{3} - i$ | (7) $4i$ | (8) $1 - \sqrt{3}i$ |

*****6.13** Using the polar form, compute the following. Express in the form $a + bi$.

- | | | |
|----------------------------|-------------------------------|----------------------------------|
| (1) $(1 + i)^8$ | (2) $((\sqrt{3} + i)/2)^{12}$ | (3) $(1 - i)^{10}$ |
| (4) $(1 + \sqrt{3}i)^{-3}$ | (5) $((1 + i)/(1 - i))^{20}$ | (6) $((\sqrt{3} + i)/(1 + i))^6$ |

*****6.14** Use de Moivre's theorem to expand $\cos(4\theta)$ and $\sin(4\theta)$ in terms of $\cos \theta$ and $\sin \theta$.

****6.15** Express $\cos^3 \theta$ and $\sin^3 \theta$ as a linear combination of $\cos(n\theta)$ and $\sin(n\theta)$ for various n .

Fail safe note: Cube the identity $\cos \theta = (e^{i\theta} + e^{-i\theta})/2$, then simplify using Eq. (6.16).

****6.16** Show the following addition formulas using Euler's formula:

(1) $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta.$

(2) $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta.$

Fail safe note: Expand $e^{i(\alpha+\beta)} = e^{i\alpha}e^{i\beta}$ and match real and imaginary parts.

Example 6.12 (Square roots of -1)

Find all square roots of -1 .

Solution

Write $w = -1 = e^{i\pi}$ (so $R = 1$, $\varphi = \pi$, $n = 2$).

$$z_0 = e^{i\pi/2} = \cos\left(\frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2}\right) = i, \quad z_1 = e^{i3\pi/2} = \cos\left(3\frac{\pi}{2}\right) + i \sin\left(3\frac{\pi}{2}\right) = -i.$$

The two square roots of -1 are i and $-i$, confirming Eq. (6.1).

Example 6.13 (Cube roots of unity)

Find all cube roots of 1 .

Solution

Write $w = 1 = e^{i \cdot 0}$ ($R = 1$, $\varphi = 0$, $n = 3$).

$$z_0 = 1, \quad z_1 = e^{2\pi i/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad z_2 = e^{4\pi i/3} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i.$$

These three points form an equilateral triangle inscribed in the unit circle.

Problems

★6.17 Find all n -th roots of the following. Draw them in the complex plane.

(1) $w = -4$, $n = 2$

(2) $w = 8i$, $n = 3$

(3) $w = -8$, $n = 3$

(4) $w = 1$, $n = 4$

(5) $w = -1$, $n = 4$

(6) $w = i$, $n = 4$

(7) $w = 1 + \sqrt{3}i$, $n = 3$

(8) $w = -\sqrt{2} + \sqrt{2}i$, $n = 2$

*****6.18** Find all solutions of the following equations in $\mathbb{C}$.

(1) $z^3 - 8 = 0$

(2) $z^4 + 16 = 0$

(3) $z^6 - 1 = 0$

(4) $z^3 + i = 0$

(5) $z^4 - 4z^2 + 3 = 0$

(6) $z^6 + z^3 - 2 = 0$

Fail safe note: For (4) and (5), try factoring or substituting $w = z^2$ or $w = z^3$.

****6.19** The n -th **roots of unity** are the n solutions of $z^n = 1$.

(1) Show that the roots of unity are ω^k for $k = 0, 1, \dots, n-1$, where $\omega = e^{2\pi i/n}$.

(2) Show that $1 + \omega + \omega^2 + \dots + \omega^{n-1} = 0$. [Hint: Consider $\sum_{k=0}^{n-1} \omega^k$ as a geometric series.]

(3) Show that the product of all n -th roots of unity is $(-1)^{n+1}$.

***6.20** Use the identity $\sum_{k=0}^{n-1} \omega^k = 0$ (from the previous problem) and de Moivre's theorem to show that for $n \geq 2$:

$$\sum_{k=0}^{n-1} \cos\left(\frac{2\pi k}{n}\right) = 0 \quad \text{and} \quad \sum_{k=0}^{n-1} \sin\left(\frac{2\pi k}{n}\right) = 0. \quad (6.17)$$

Chapter 7

Matrices